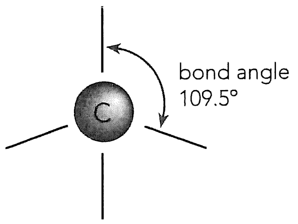
4.1 INTRODUCTION TO BONDING AND TERMINOLOGY

Carbon has the electron structure **2 4 ($1s^2$ $2s^22p^2$)**. To obtain the same electron configuration as the nearest noble gas, neon; carbon can form four covalent bonds. Carbon has the unique ability to form *very stable (strong) covalent bonds* with itself and other elements such as hydrogen, oxygen, nitrogen and the halogens. The stability of the bonds allows carbon to form a virtually unlimited number of straight chain, branched chain and ring molecules.

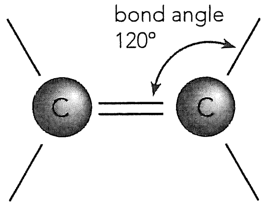
The bonding options for carbon atoms are:

1. four single covalent bonds with a tetrahedral structure as the basic shape.



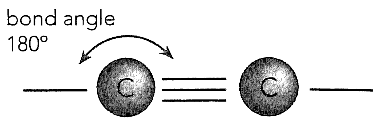
An **alkane** is a carbon chain that contains only single bonds between carbon atoms. Alkanes have the general formula C_nH_{2n+2}

2. one double bond and two single bonds with a triangular planar structure as the basic shape.



An **alkene** is a carbon chain that contains at least one double bond between carbon atoms. Alkenes have the general formula C_nH_{2n} (also cycloalkanes)

3. one triple bond and one single bond with a linear structure as the basic shape.

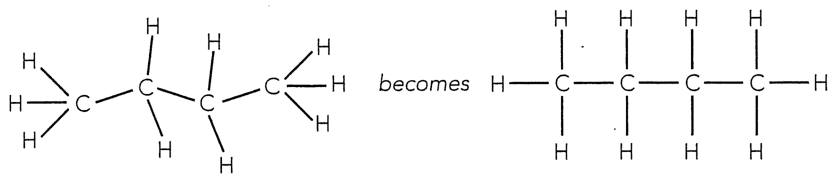


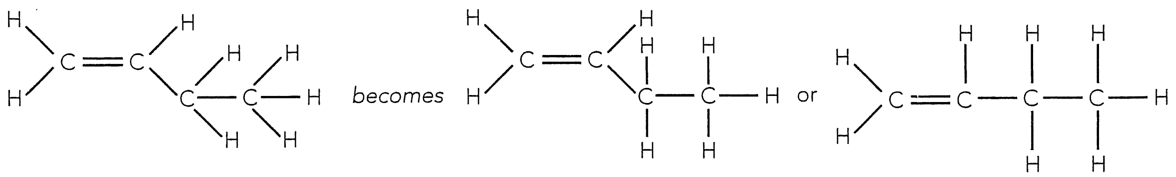
An **alkyne** is a carbon chain that contains at least one triple bond between carbon atoms. Alkynes have the general formula C_nH_{2n-2} (also cycloalkenes)

- Carbon often forms single bonds with *other carbon atoms, hydrogen, oxygen, nitrogen and the halogens.*
- Carbon often forms double bonds with *other carbon atoms and oxygen.*
- Carbon often forms triple bonds with *other carbon atoms and nitrogen.*

Drawing Organic Molecules

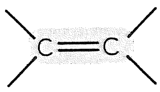
Most organic molecules are based on a tetrahedral shape in which the bonding electron pairs assume a maximum angle of separation in 3-dimensional space. In tetrahedral molecules the bond angles are 109.5°. Representing 3-dimensional figures on 2-dimensional paper is sometimes confusing, the general procedure followed in drawing the 3-D tetrahedral shape on 2-D paper is to draw the bond angles as 90°. Two examples are given below:





Functional Groups

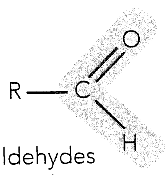
These are the chemically reactive part of an organic molecule. Some important functional groups are:



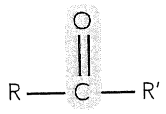
alkenes
(-ene)

R — OH

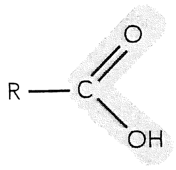
alchols
(-ol)



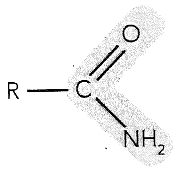
aldehydes
(-al)



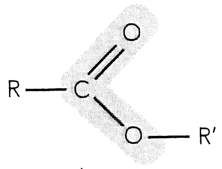
ketones
(-one)



carboxylic acids
(-oic)



amides
(-amide)



esters
(-yl-oate)

R— NH_2

amines
(-amine)

Alkyl Groups

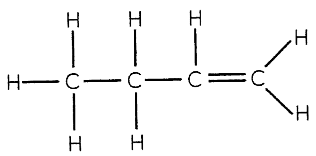
These are saturated carbon chains (what does saturated mean?) which, together with functional groups, compose organic molecules.

Alkyl groups are hydrocarbon chains that are missing one hydrogen. The place where the hydrogen atom is missing is where the alkyl group is attached to the main carbon chain. Alkyl groups have the general formula C_nH_{2n+1} and are often indicated by the symbols R, R_1, R_2, etc.

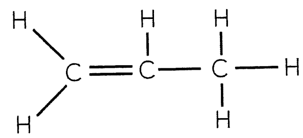
Question 4.1

Examine the following diagrams (structural formulae) and identify the functional **group** present.

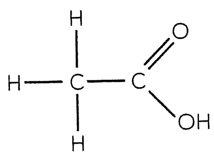
(a)



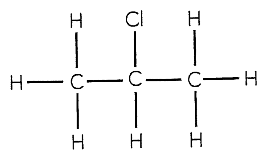
(b)



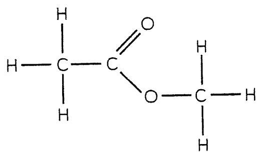
(c)



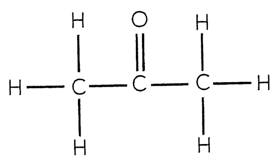
(d)



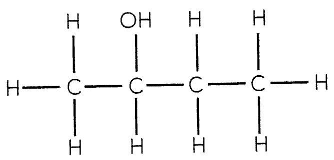
(e)



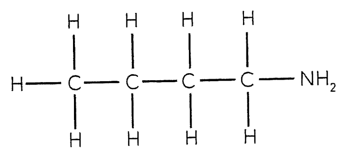
(f)



(g)



(h)



IUPAC rules for naming organic molecules

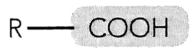
These rules are open to some degree of interpretation and so there will always be some disparity between names given by different chemists.

A small selection of the IUPAC Rules are:

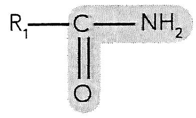
1. Identify the functional groups in the molecule and determine which group has the highest priority in naming (order of priorities given below).
2. Select the longest continuous carbon chain containing the highest priority functional group.
3. Number the chain from the end which gives the main functional group the lowest number.
4. Number all functional groups using the order from #3.
5. Use the appropriate prefix or suffix to name each functional group.
6. Use one word to name the compound. The name for each group is preceded by a numeral indicating its attachment position in the parent chain. Alphabetical order is used when more than one group is involved.
7. Numerals are separated from words by a hyphen and from other numerals by a comma.
8. The priorities for selecting functional groups (R– is a carbon chain of zero or more carbon atoms in length):

decreasing priority

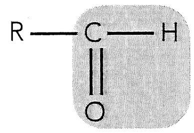
(i) carboxylic acids



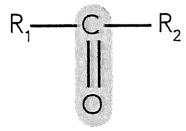
(ii) amides



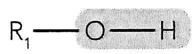
(iii) aldehydes



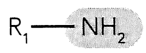
(iv) ketones



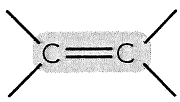
(v) alcohols



(vi) amines



(vii) alkenes



(viii) alkyl groups, C_6H_5 - , halides, etc

Question 4.2

Complete the following table to show the prefixes that are used for indicating the number of carbon atoms in a chain.

Number of carbon in chain	alkane	alkene (with 1 double bond only)	alkyl group
1	methane	***	methyl
2	ethane	ethene	ethyl
3	propane		
4	butane		
5	pentane		
6	hexane		
7	heptane		
8	octane		

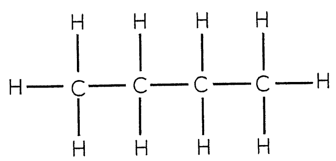
When naming molecules that have more than one of a functional group attached the following prefixes are used:

2	3	4	5	6	7	8	9	10
di	*tri*	*tetra*	*penta*	*hexa*	*hepta*	*octa*	*nona*	*deca*

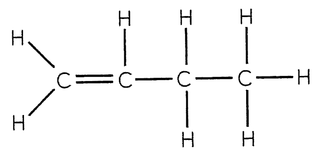
Question 4.3

Use IUPAC Rules to name the following organic compounds.

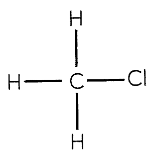
(a)



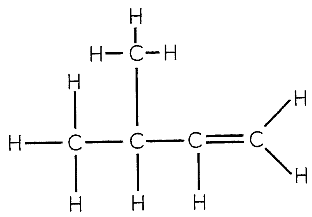
(b)



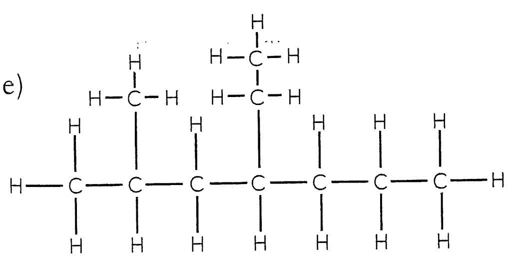
(c)



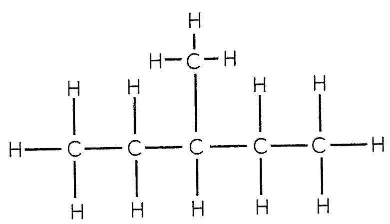
(d)



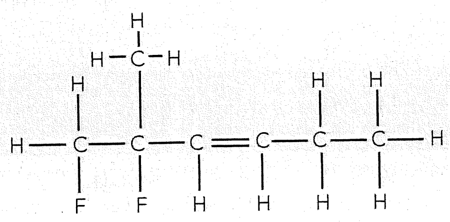
(e)



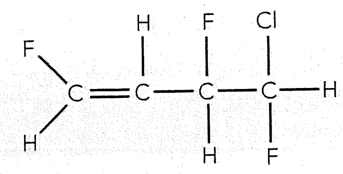
(f)



(g)



(h)



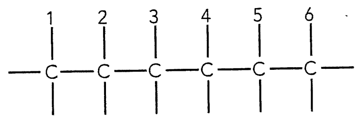
Structural Formula

A structural formula diagram gives a 2-dimensional representation of the organic molecule. It provides important information about where each different atom is attached to a carbon chain. A molecular formula provides the correct number of atoms of each element in a compound, a structural formula provides that **plus** the bonding position of each atom of each element.

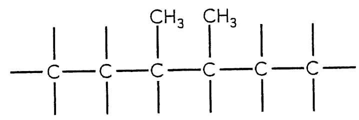
Worked Example 4.1

Draw the structural formula for 3,4-dimethylhexane.

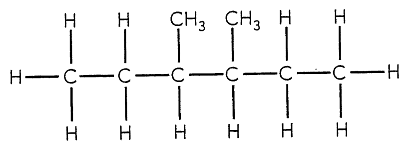
(i) Identify the longest chain by reading the end of the name - hexane.



(ii) Identify what is attached to the longest chain and where it is attached - two methyls at carbons numbered 3 & 4.



(iii) Complete the diagram by drawing the remaining hydrogens.



Question 4.4

Draw the structural formulae of the following compounds.

(a) 3,4-diethyloctane

(b) 1,3,4-triiodohexane

(c) 1,1-dichloropent-2-ene

(d) 3,6-diethylnon-4-ene

Question 4.5

Identify what is **incorrect** about the name of each of the following.

(a) but-3-ene ______________________________

(b) 2-ethylprop-1-ene ______________________________

(c) pent-4-ene ______________________________

(d) 3,4-dimethylpent-4-ene ______________________________

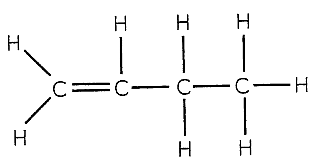
4.2 ISOMERISM

Structural isomers

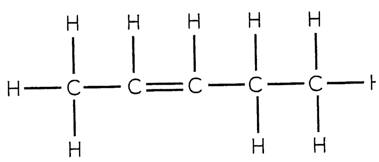
Structural isomers are molecules that have the same molecular formula but have different properties. They have different chemical and physical properties because they have different arrangements of the atoms in the molecule, i.e. they have different structures.

Some examples of structural isomers:

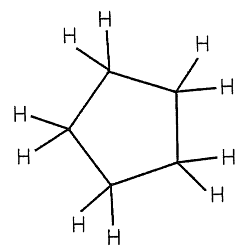
Molecular formula: C_5H_{10}



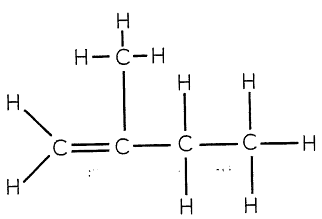
isomer 1: pent-1-ene



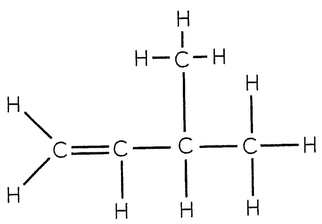
isomer 2: pent-2-ene



isomer 3: cyclopentane



isomer 4: 2-methylbut-1-ene



isomer 5: 3-methylbut-1-ene

Question 4.6

Draw the structural formula for each of the following molecules. They all have the molecular formula C_5H_{10}, so identify which are structural isomers of those drawn above. For those that are not isomers – identify why not.

(a) 2-methylbut-2-ene

(b) 3-methylbut-2-ene

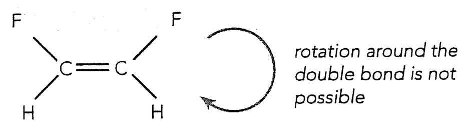
(c) methylcyclobutane

(d) pent-3-ene

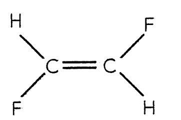
Cis and trans isomers

Cis and trans isomers occur only in alkenes. The existence of the double bond stops rotation and so molecules with the same structural formula can have different geometries and hence different properties.

e.g. 1,2-difluoroethene has two isomers:



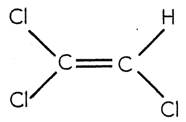
cis-1,2-difluoroethene



trans-1,2-difluoroethene

The cis prefix refers to the different structures being on the same side of the axis splitting the double bond while the trans prefix refers to the different structures being across the axis splitting the double bond.

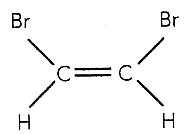
Not all alkene compounds exhibit cis/trans isomerism, e.g.



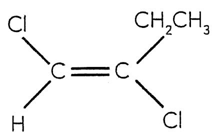
Worked Example 4.2

Name the following cis/trans isomers.

(a)



(b)



STEP 1: Name the compound as per normal.

STEP 2: Identify both examples, identify the species that are on the same side (*cis*) or opposite sides (*trans*) of the double bond.

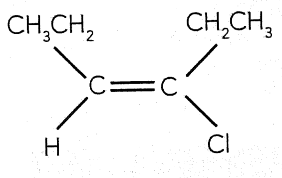
STEP 3: Place the appropriate suffix (*cis* or *trans*) in front of the name.

(a) *cis*-1,2-dibromoethene (b) *trans*-1,2-dichlorobut-1-ene

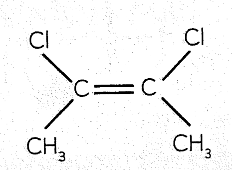
Question 4.7

Name the following compounds using the prefixes "cis" and "trans".

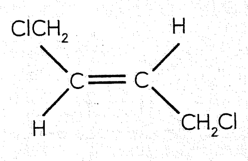
(a)



(b)



(c)



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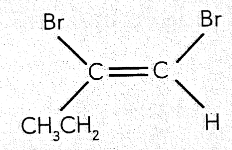


Question 4.8

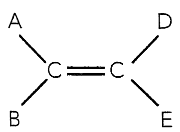
Draw the *cis* and *trans* isomers for each of the following (if the isomers exist).

(a) hept-1-ene (b) 1,2-dichloroethene (c) 1-chloropent-1-ene

SUMMARY – ISOMERS

Structural isomers have the same molecular formula but have:

- different carbon configurations,
- the same functional group in different positions on the chain,
- the same carbon chain with different functional groups.



Cis/trans isomers must be alkenes and have the general form:

where A ≠ B and D ≠ E and C is a carbon atom.

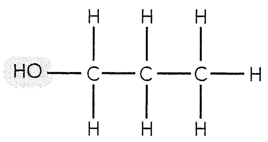
4.3 FUNCTIONAL GROUPS

Alcohols

General formula: -C-OH, or R-OH.

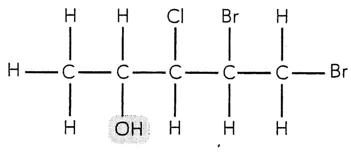
Naming: the compound's name will end in -ol.

e.g. 1:



is propan-1-ol

e.g. 2:

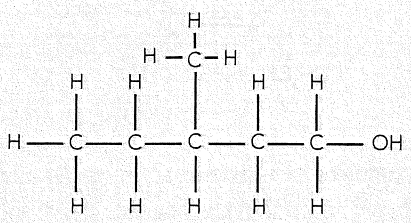


is 4, 5-dibromo, 3-chloro-pentan-2-ol

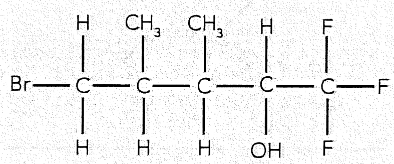
Question 4.9

Name the following compounds.

(a)

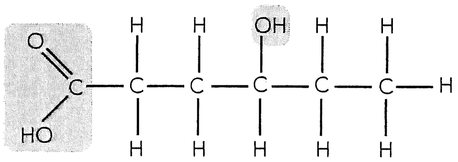


(b)



Note: Using **-ol** as a suffix will not occur when a functional group of higher priority is attached to the carbon chain. In this case the presence of the -OH functional group is indicated using the prefix **-hydroxy-**.

e.g.



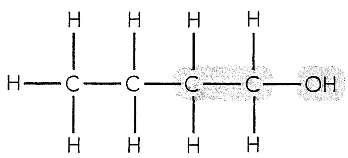
is 4-hydroxyhexanoic acid

Primary, Secondary and Tertiary Alcohols

Alcohols can also be identified according to how many carbon atoms are bonded to the carbon atom attached to the hydroxy group.

Primary alcohol: – OH is attached to a carbon which is attached to **one** other carbon atom.

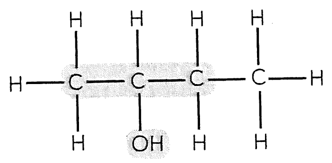
e.g.



butan-1-ol

Secondary alcohol: – OH is attached to a carbon which is attached to **two** other carbon atoms.

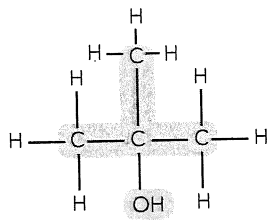
e.g.



butan-2-ol

Tertiary alcohol: – OH is attached to a carbon which is attached to **three** other carbon atoms.

e.g.



methylpropan-2-ol

Reactions of alcohols

1. oxidation of primary alcohol → aldehyde → carboxylic acid

e.g. Write the half equations and full equation for the oxidation of propan-1-ol by acidified potassium permanganate to form propanal.

Oxidation: $CH_3CH_2CH_2OH \rightarrow CH_3CH_2C(=O)H + 2H^+ + 2e^-$ (× 5)

Reduction: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (× 2)

Nett: $5CH_3CH_2CH_2OH + 2MnO_4^- + 6H^+ \rightarrow 5CH_3CH_2C(=O)H + 2Mn^{2+} + 8H_2O$

NB: propanal could be oxidised further to form propanoic acid.

$$CH_3CH_2C(=O)H + H_2O \rightarrow CH_3CH_2C(=O)OH + 2H^+ + 2e^-$$

2. oxidation of a secondary alcohol → ketone

e.g. Write the half equations and nett equation for the oxidation of propan-2-ol by acidified potassium dichromate.

Oxidation: $CH_3\overset{OH}{\overset{|}{C}}HCH_3 \rightarrow CH_3\overset{O}{\overset{\|}{C}}CH_3 + 2H^+ + 2e^-$ (× 3)

Reduction: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

Nett: $3CH_3\overset{OH}{\overset{|}{C}}HCH_3 + Cr_2O_7^{2-} + 8H^+ \rightarrow 3CH_3\overset{O}{\overset{\|}{C}}CH_3 + 2Cr^{3+} + 7H_2O$

3. NB: tertiary alcohols are not oxidised by acidified $KMnO_4$, etc

4. alcohol + carboxylic acid $\xrightarrow{H^+}$ ester + water
See section on esters for more detail.

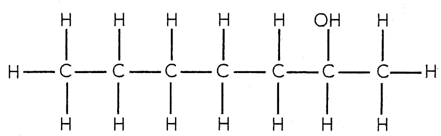
Properties of alcohols

Due to the presence of the -OH group, alcohols exhibit hydrogen bonding. This means that alcohols will tend to have higher melting points and boiling points than other hydrocarbons of similar molar mass. The hydrogen bonding leads to alcohols having a relatively high solubility in water – although the solubility decreases as the length of the hydrocarbon chain increases.

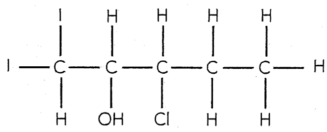
Because of the openness of the hydrocarbon chain primary alcohols tend to have higher boiling points than secondary alcohols which tend to have higher boiling points than tertiary alcohols (this only applies to alcohols of similar molar mass).

Question 4.10

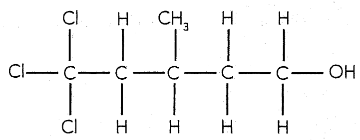
Name the following organic compounds.

(a) 

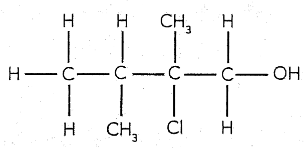
____

(b) 

____

(c) 

____

(d) 

____

Question 4.11

Draw the structural formula for each of the following.

(a) 2-iodopropan-2-ol

(b) 4,5-difluoro-6-methyloctan-2-ol

Question 4.12

(a) Use half equations to write the balanced equation for acidified $K_2Cr_2O_7$ oxidising butan-1-ol to form butanal.

Oxidation: ____________________________

Reduction: ____________________________

(b) Use half equations to write the balanced equation for the reaction between acidified $KMnO_4$ and pentan-3-ol.

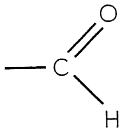
Oxidation: ____________________________

Reduction: ____________________________

(c) Write the equation for the complete oxidation of butan-1-ol by oxygen gas.

____

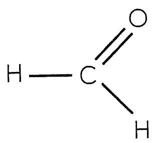
Aldehydes

General formula:  or – CHO

Naming: aldehydes are named using the suffix **-al** in place of the "e" on the end of the hydrocarbon's name.

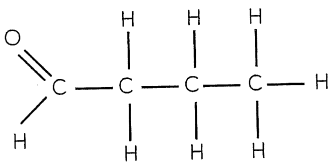
Aldehydes (and ketones) contain the –C=O functional group which is named the carbonyl group. The presence of this carbonyl group causes aldehydes to exhibit dipole-dipole interactions. This gives aldehydes higher melting points, boiling points and solubilities in water than other hydrocarbons of similar size. (These properties are not as pronounced or magnified as in alcohols which exhibit hydrogen bonding.)

e.g. 1



methanal (formaldehyde)

e.g. 2



butanal

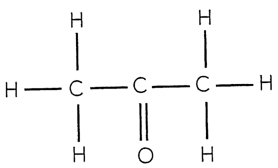
As previously discussed, aldehydes are formed by the oxidation of primary alcohols. Aldehydes can be oxidised to form carboxylic acids.

Ketones

General formula: $-\overset{O}{\overset{\|}{C}}-$ or $R_1-\overset{O}{\overset{\|}{C}}-R_2$

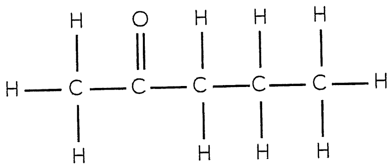
Naming: ketones are named using the suffix **-one** in place of the "e" on the end of the hydrocarbon's name.

e.g. 1



propanone (acetone)

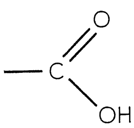
e.g. 2



pentan-2-one

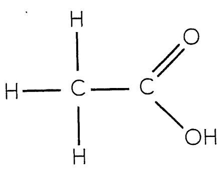
The properties of ketones are very similar to those of aldehydes. However, unlike aldehydes, ketones are not oxidised to form carboxylic acids. Ketones are formed by the oxidation of secondary alcohols.

Carboxylic acids

General formula: –COOH or  or RCOOH

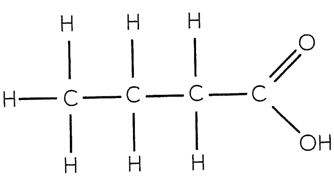
Naming: carboxylic acids are named using the suffix **-oic acid** in place of the "e" on the end of the hydrocarbon's name.

e.g. 1



ethanoic acid

e.g. 2



butanoic acid

NB: aldehydes and carboxylic acids must be at the end of the carbon chain and so are assumed to be number 'one' in a name.

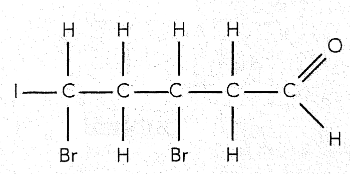
Because of the presence of the –C=O and –OH groups carboxylic acids have high solubilities in water, melting points and boiling points when compared to other organic compounds of similar molar mass. As with alcohols, aldehydes and ketones; the solubility of carboxylic acids decreases as the length of the carbon chain increases.

Carboxylic acids are weak acids as they do not completely ionise when dissolved in water. Carboxylic acids are formed by the oxidation of primary alcohols and aldehydes.

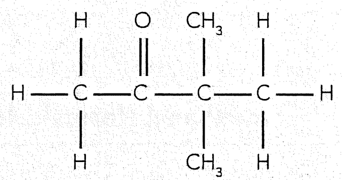
Question 4.13

Name the following compounds.

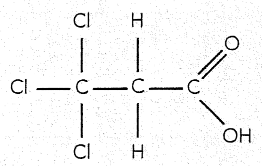
(a)



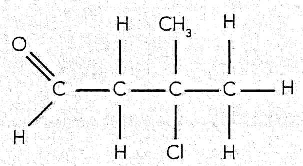
(b)



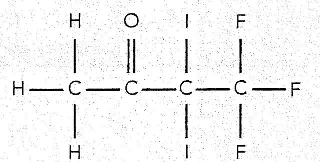
(c)



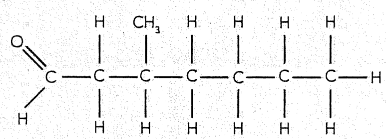
(d)



(e)



(f)



Question 4.14

Draw the structural formula for each of the following.

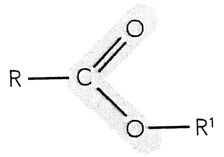
(a) 3,4-dichloropentanal

(b) 3-methyloctan-4-one

(c) hexanoic acid

(d) 3,3,3-triiodopropanal

Esters

General formula: 

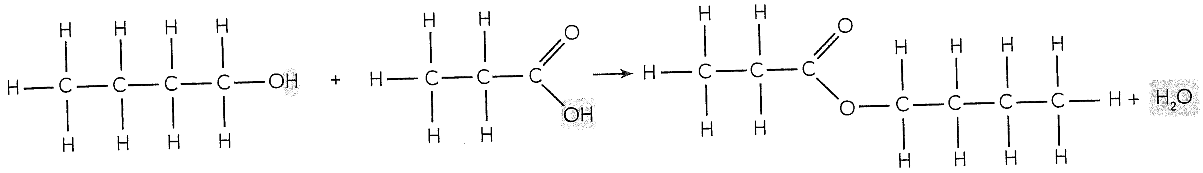
Naming: esters are made by reacting a carboxylic acid with an alcohol. Their name is derived from the two reacting components.

e.g. 1 ethanol + **methanoic acid** $\xrightarrow{H^+\text{(catalyst)}}$ ethyl methano**ate** + water
where the **yl** comes from the alcohol reactant and the **oate** comes from the acid reactant.

Esterification is the name for the chemical reaction in which esters are made.

e.g. 2 Write the equation for the reaction between butan-1-ol and propanoic acid.

butan-1-**ol** + propanoic **acid** $\xrightarrow{H^+}$ butyl propanoate + water



Question 4.15

Name the ester formed in each of the following, and draw its structural formula.

(a) methanol + propanoic acid $\xrightarrow{H^+}$

(b) hexanoic acid + propan-1-ol $\xrightarrow{H^+}$

(c) heptan-1-ol + ethanoic acid $\xrightarrow{H^+}$

Question 4.16

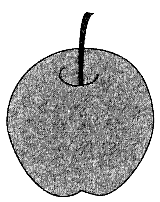
What organic reactants could be used to produce the following esters?

(a) ethyl propanoate ____________________

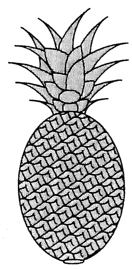
(b) octyl butanoate ____________________

(c) propyl methanoate ____________________

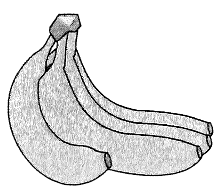
Properties of esters: Esters have similar boiling points to aldehydes and ketones with the same number of C atoms because of dipole-dipole force of attraction and dispersion forces. Small esters are soluble in water but the solubility decreases as the length of the carbon chain increases. Many esters have their own distinctive fruity smell.



methyl butanoate

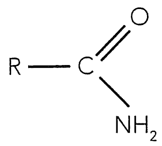


ethyl butanoate

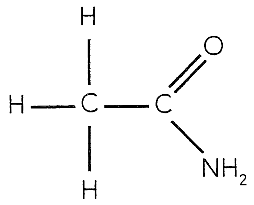


pentyl ethanoate

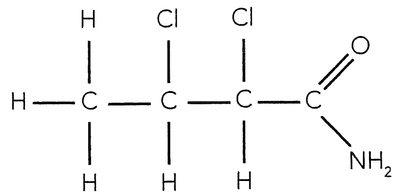
Amides

General Formula: 

Naming: amides are named using the suffix **-amide** in the place of the "e" on the end of the hydrocarbon's name.

e.g. 1: 

ethanamide

e.g. 2: 

2,3-dichlorobutanamide

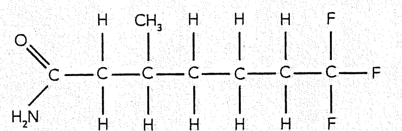
Amides can be produced by reacting ammonia with a carboxylic acid. The product is then heated to remove the water.

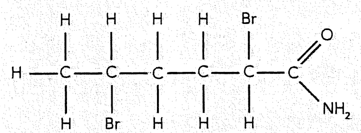
$$RCOOH + NH_3 \xrightarrow{\text{Heat}} RC=ONH_2 + H_2O$$

The presence of the NH_2 and C=O groups mean that amides form strong hydrogen bonds and consequently have high melting points for their size and are soluble in water. As is usual with organic molecules the solubility decreases as the size of the molecule increases. Although the presence of the -NH_2 would suggest otherwise, amides tend to form neutral solutions.

Question 4.17

Name the following compounds.

(a) 

(b) 

___________________________ ___________________________

Question 4.18

Draw the structural formula of the following

(a) 4,5,6-triiodoheptanamide

(b) 3-chloro-4-methylpentanamide

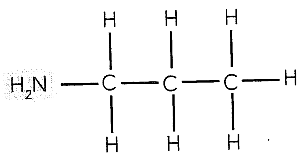
4.4 AMINES AND α-AMINO ACIDS

Amines

General formula: $-NH_2$ or $R-NH_2$.

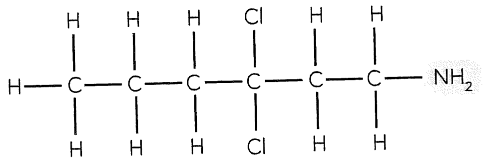
Naming: the compound's name will end in –amine.

e.g. 1



propan-1-amine

e.g. 2



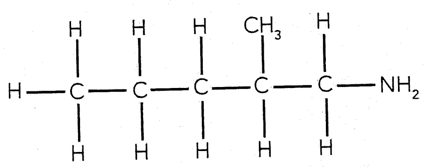
3, 3-dichlorohexan-1-amine

Note: **–amine** as a suffix is not used when a functional group of higher priority is attached to the carbon chain. In this case the presence of the **$-NH_2$** functional group is indicated using the prefix **–amino-**

Question 4.19

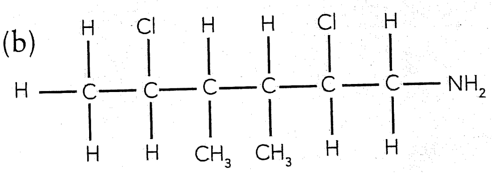
Name the following compounds:

(a)



____

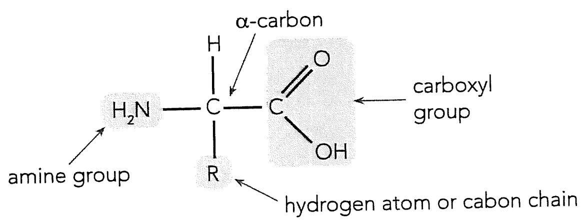
(b)



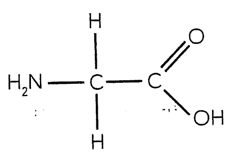
____

α-Amino Acids

These compounds are the building blocks of proteins. They have an amine and a carboxyl group attached to the same carbon. This carbon is called the alpha (α) carbon.

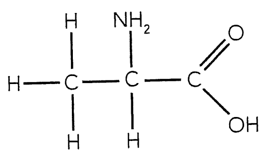


e.g. 1



glycine (2-amino ethanoic acid)

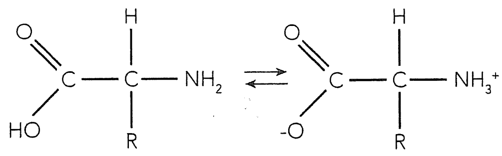
e.g. 2



alanine (2-amino propanoic acid)

Zwitterion formation: under some circumstances the amine group on the α-amino acid removes a proton (H^+) from the carboxylic group on the α-amino acid and the α-amino acid becomes a neutral ion that has a positive region (NH_3^+) and a negative region (COO-). This neutral ion that possesses both a positive and a negative charge is called a **zwitterion.**

e.g.

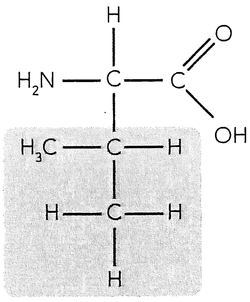


The properties of α-amino acids determine the properties of the proteins that they make. Generally it is the properties of the carbon chain (R) attached to the α-carbon that determine the properties of the α-amino-acid. The main properties are the acidic/basic nature of the α-amino acid and its polarity.

The polarity is linked to its solubility in water.

CASE 1: R is a hydrocarbon, it will be neutral and non polar

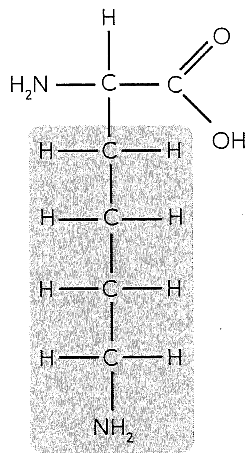
e.g. **valine:**



valine considered to be neutral and non polar

CASE 2: R has an amine group at the end, it will be basic and polar

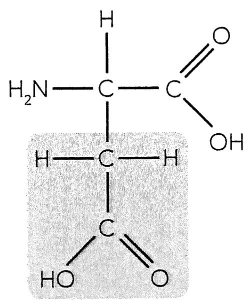
e.g. **lysine:**



the $-NH_2$ at the end of the chain makes lysine basic and polar

CASE 3: R has a carboxylic acid group at the end, it will be acidic and polar

e.g. **aspartic acid**



the –COOH at the end of the chain makes aspartic acid acidic and polar

NB: Alcohol groups at the end of "R" will make the α-amino acid polar and neutral, e.g. serine.

Question 4.20

Glycine is the simplest amino acid. Comment on its polarity and acidity.

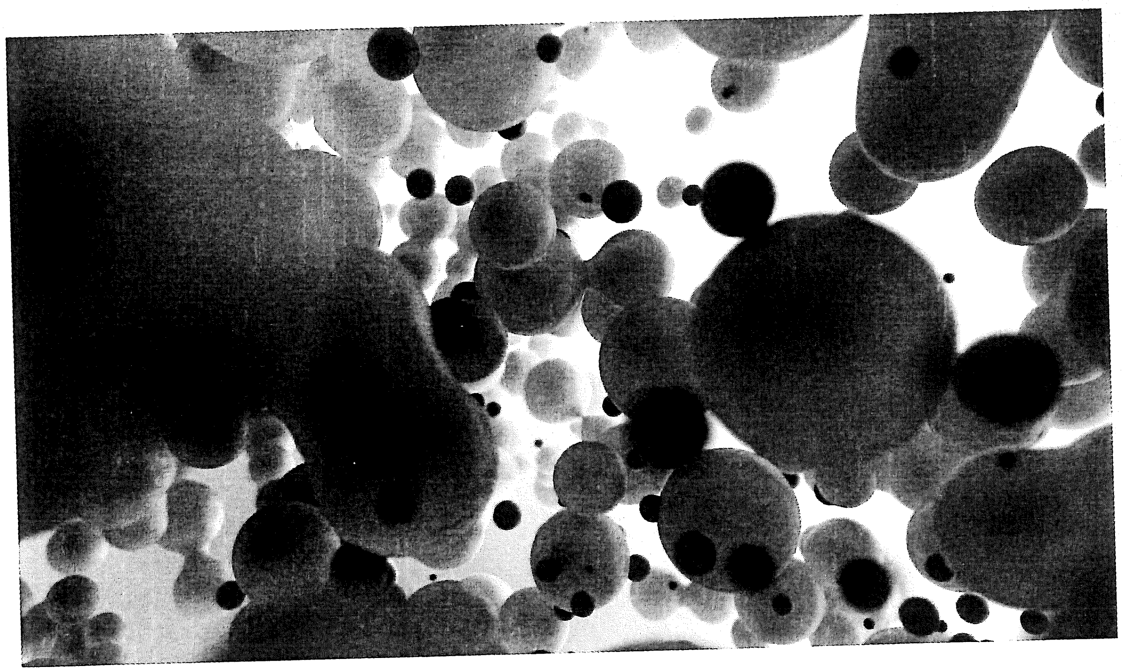
Question 4.21

Write the equation for glycine dissolving in:

(a) water

(b) NaOH solution

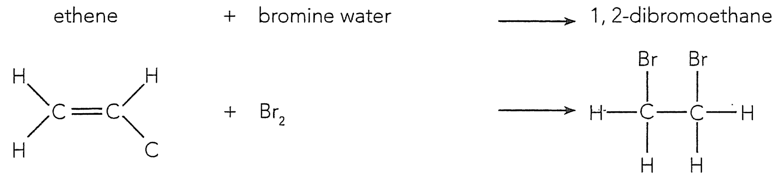
(c) HCl solution



4.5 REACTIONS OF HYDROCARBONS

Addition reactions

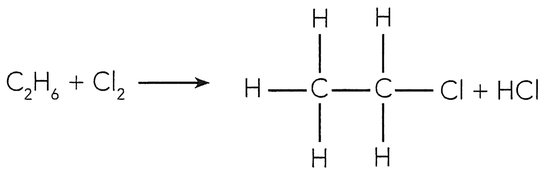
Addition reactions occur when **alkenes** and **alkynes** have their double or triple bond broken and elements such as halogens added into the carbon chain.



Substitution reactions

Substitution reactions occur when alkanes and aromatic compounds react such that a hydrogen atom is removed and another element such as a halogen is substituted into its position on the hydrocarbon chain.

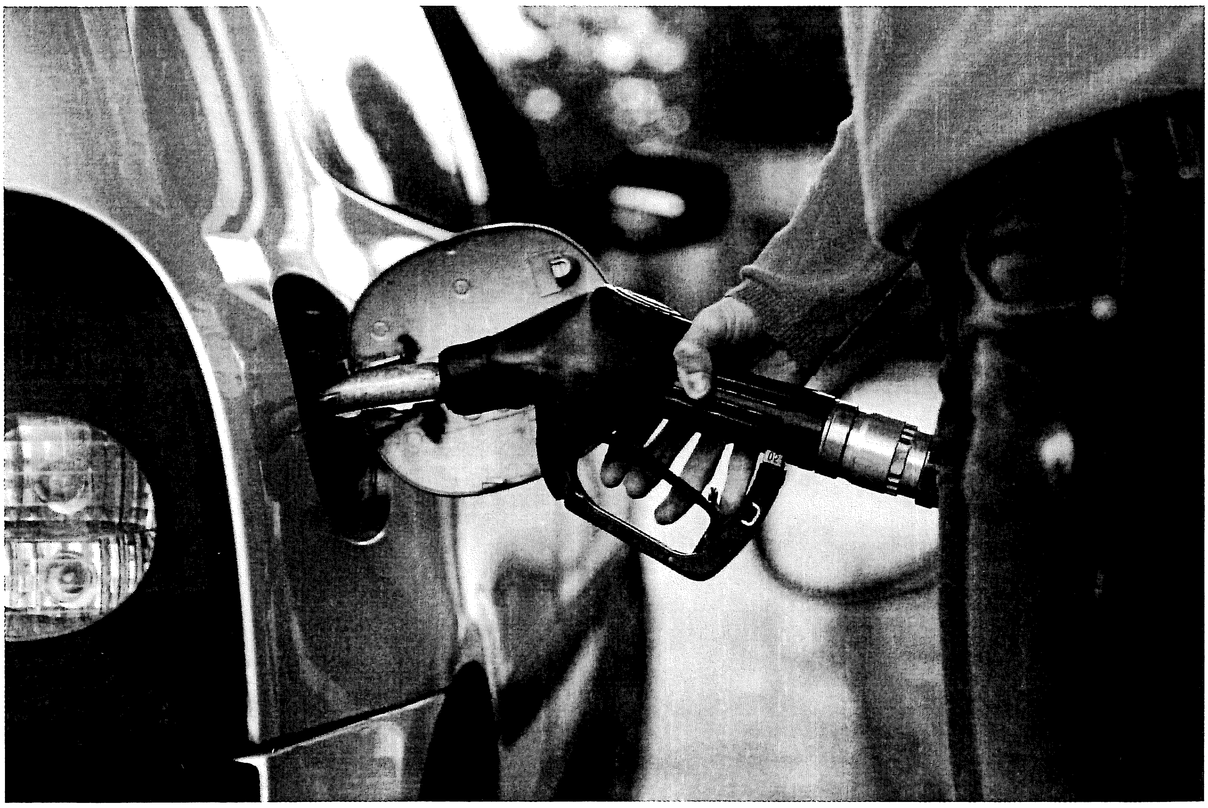
e.g.



Alkanes and aromatic compounds are relatively inert compounds. Substitution reactions need to occur at moderately high temperatures (250°C) or with UV light added and are slower than addition reactions (typically).

Combustion

All hydrocarbon compounds can be burnt in oxygen. If complete combustion occurs then the only products are carbon dioxide and water vapour. The production of energy by burning organic fuels (fossil fuels) is the major form of energy production on Earth.



Question 4.22

Write the equation for the following reactions. Show the structural formula of all organic compounds.

(a) ethene + Cl_2 ______________________________

(b) hexane + Br_2 ______________________________

(c) but-1-ene + HBr ______________________________

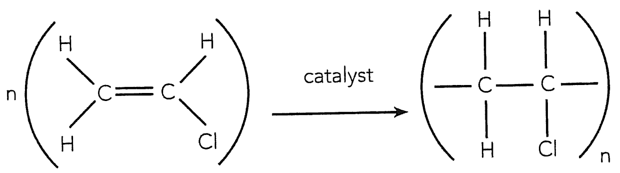
(d) C_7H_{16} + O_2 ______________________________

4.6 POLYMERISATION

A **monomer** is a small molecule that can be joined many times to form a polymer. **Polymerisation** is the chemical process by which many monomers are linked to each other to form large chain molecules called polymers.

Two types of polymerisation are:

i) **Addition Polymerisation:** the monomer is an alkene and the polymer is formed by the breaking of the double bond in the simple alkene and then linking the pieces together.



vinyl choride (chloroethene) polyvinyl choride

ii) **Condensation Polymerisation:** one monomer is a di-carboxylic acid and the other is either a di-alcohol or a di-amine. If the non-acidic monomer is a di-alcohol the polymer formed will be a polyester but if it is a di-amine then the polymer is a polyamide. In either case the other product will be water and hence the name condensation polymer.

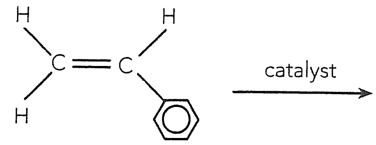
n(HOOC–R–COOH)	+	n(HO–R´–OH)
–dioic acid	+	–diol
→ (–OC–R–COOR´O–)n	+	$2H_2O$
polyester	+	water
n(HOOC–R–COOH)	+	$n\left(\begin{matrix}H\\H\end{matrix}\!>N-R'-N<\!\begin{matrix}H\\H\end{matrix}\right)$
–dioic acid	+	–diamine
→ (–OC–R–CONHR´NH–)n	+	$2H_2O$
polyamide	+	water

Wool, silk, nylon and Kevlar™ are examples of polyamides while terylene and polyethylene terephthalate (PET) are examples of polyesters.

Question 4.23

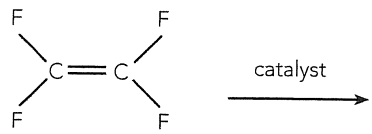
Show what polymers can be formed from the following monomers.

(a)



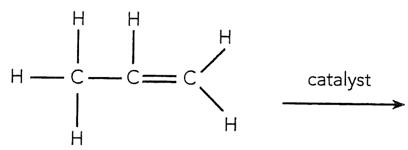
monomer = styrene polymer = ____________________

(b)



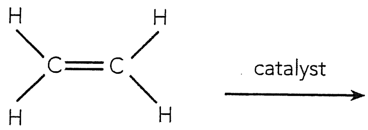
monomer = tetrafluoroethene polymer = teflon

(c)



monomer = propene polymer = ____________________

(d)

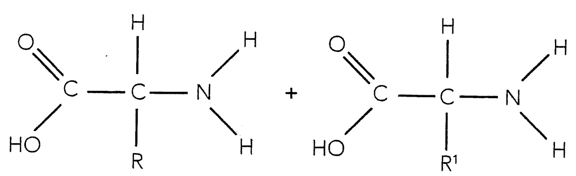


monomer = ethene polymer = ____________________

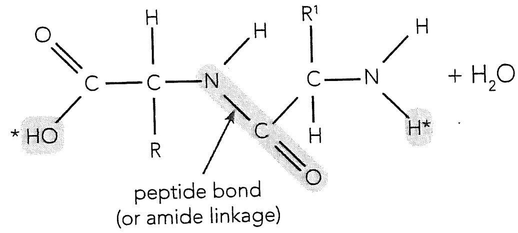
Polypeptides and proteins

Polypeptides are chains of α-amino acids. Proteins are made up of one or more polypeptide molecules. Proteins are naturally occurring condensation polymers that have α-amino acids joined together in a vast arrangement of different sequences. The mechanism by which the α-amino acids polymerise is:

1. A hydrogen atom is removed from the amine group from one a-amino acid molecule and an OH group is removed from the carboxyl end of another a-amino acid molecule.



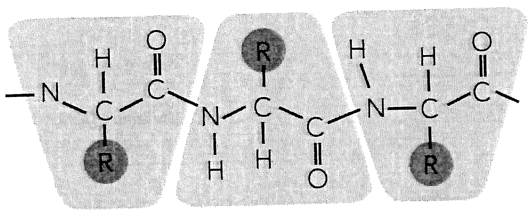
2. The two molecules then join at the sites where the H and OH are missing in what is called a peptide bond.



3. This also occurs at the other ends of the α-amino acid molecules as indicated by the asterisk in (2). When many α-amino molecules join together, the polymer is called a polypeptide.

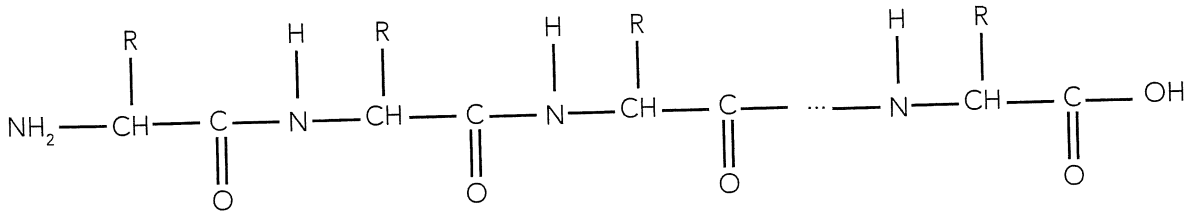
4. One polypeptide molecule can join or bond to another polypeptide molecule because of interactions (dispersion forces, dipole-dipole interactions or hydrogen bonds) between the side chain section labelled R and R_1 in the diagrams of points (1) and (2).

5. The amide linkage, or peptide bond, is the key structure to identify when describing proteins. It is usual to start at the amine (-NH-) end and go to the carbonyl end (-C=O). The diagram below shows the repetition of a small number of amino acid monomers in a protein.



The sequence of the repeating amino acid monomers gives the protein its **primary** structure. The **primary** structure of a protein relates to structure governed by the covalently bonded atoms in the protein but not including any disulfide bridges.

The diagram below shows a common representation for the primary structure of a protein.



A protein may contain between 50 and 2000 repetitions of the amino acid monomers. The "R" side chain represents the remaining carbon chain region (not the NHCHC=O section) of any of the 20 amino acids that make up proteins. It is common practice to use a 3-letter or a 1-letter abbreviation for the amino acid rather than "R" when drawing proteins. The polarity of the side chain tends to affect the interaction of that region of the molecule with water. Generally the non-polar side chain regions of the protein tend to be at the core of the protein whereas the polar or charged regions tend to be at the surface.

Table 4.1

Protein	Abbreviation		Protein	Abbreviation	
Glycine (non-polar)	Gly	G	Threonine (polar)	Thr	T
Alanine (non-polar)	Ala	A	Cysteine (polar)	Cys	C
Valine (non-polar)	Val	V	Methionine (polar)	Met	M
Leucine (non-polar)	Leu	L	Asparagine (polar)	Asn	N
Isoleucine (non-polar)	Ile	I	Glutamine (polar)	Gln	Q
Proline (non-polar)	Pro	P	Lysine (+vely charged)	Lys	K
Phenylalanine (non-polar)	Phe	F	Arginine (+vely charged)	Arg	R
Tyrosine (polar)	Tyr	Y	Histidine (+vely charged)	Hi	H
Tryptophan (polar)	Trp	W	Aspartate (-vely charged)	Asp	D
Serine (polar)	Ser	S	Glutamate (-vely charged)	Glu	E

Question 4.24

How many monomer units are drawn in the protein on the previous page? __________

Primary bonding within between atoms in a protein chain and secondary bonding between charged regions within protein chains affect the shape and properties of protein molecules.

Table 4.2

Primary Structure	Covalent bonding between atoms within the protein
Secondary Structure (α-helix)	Hydrogen bonding between amide and carbonyl groups within a single chain. This tends to pull the protein chain into a coiled, helix structure.
Secondary Structure (β -pleated sheets)	Hydrogen bonding between amide and carbonyl groups in adjacent chains. This will tend to cause folds or pleats in the protein chains
Tertiary Structure	Bonding between "R" region or side chains that contribute to overall 3-D structure of the protein. The bonding between the side chains may be hydrogen bonding, dipole-dipole interactions, dispersion forces, ionic interactions or disulfide bridges. (See table 4.1)

4.7 ORGANIC SOLVENTS

The solubility of one substance in another is difficult to predict and is dependent on factors such as the nature of intermolecular forces within the solvent (and solute), intermolecular forces between solvent and solute molecules, quantities of solvent and solute, temperature, etc.

A **general** guide to the solvent qualities of organic substances is:

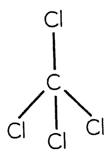
Polar solutes dissolve in *polar* solvents,
non-polar solutes dissolve in *non-polar* solutes,
BUT
non-polar solutes do not dissolve in *polar solvents,*
polar solutes do not dissolve in *non-polar solvents.*

For organic compounds the following table may be useful:

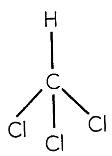
Polar Solvents	Non-polar Solvents
alcohols (–OH)	alkanes
amines (–NH_2)	alkenes
aldehydes (–CHO)	alkynes
ketones (–C=O)	esters
carboxylic acids (–COOH)	

- As the length of the hydrocarbon chain increases, the molecules listed as polar solvents tend to become less polar and more non-polar in nature.
- The polar nature of alkyl halides is not predictable and needs to be determined through examining the shape and dipole moment or overall polarity of the molecules involved. Alkyl halides that are polar tend not to have as strong a polar nature as water and are often only slightly soluble in water.

e.g.



CCl_4 contains polar covalent bonds because of its shape is non-polar.



$CHCl_3$ is polar but is only slightly soluble.

Petrol, kerosene, carbon tetrachloride, mineral turpentine and some esters are common non-polar solvents.

Ethanol, acetone (propanone) and methylated spirits are common polar solvents.

Question 4.25

Explain how a small amount of motor oil could be removed from a concrete driveway.

Question 4.26

Methylated spirits is useful for cleaning windows. Explain what sort of materials methylated spirits would have trouble removing and explain why it doesn't leave streaks on the window.

____

____

____

____

Question 4.27

Complete the following table.

Functional group	General formula	Name ends in:	Solubility in water	Produced by: (give an example of a typical reaction)
–Cl –I –Br (haloalkane)				substitution addition
>C=C< (alkene)				NA
$-NH_2$ (primary amine)				NA
–OH (alcohols)				hydration of an
–CHO (aldehydes)				oxidation of a
$-\overset{\underset{\|\|}{O}}{C}-$ (ketones)				oxidation of a
–COOH (carboxylic acids)				oxidation of a or of an
$-\overset{\underset{\|\|}{O}}{C}-O-$ (esters)				alcohol with

Question 4.28

Complete the following table.

Reactants	Organic Products	Observations
Primary alcohol + $KMnO_4$ (acidified)		
Secondary alcohol + $K_2Cr_2O_7$ (acidified)		
Tertiary alcohol + $KMnO_4$ (acidified)		
Alkane + Cl_2 (with UV light)		
Alkyne + Cl_2		

Question 4.29

Complete the following organic reactions.

Combustion

$$\left[\begin{array}{l}\text{Hydrocarbons}\\\text{Alcohols}\\\text{Aldehydes, ketones}\\\text{Esters}\\\text{Carboxylic acids}\end{array}\right] + O_2 \rightarrow \underline{\hspace{3cm}} + \underline{\hspace{3cm}}$$

Substitution and Addition

Alkane + Halogen → _______________ + _______________

Alkene + Halogen → _______________

Oxidation of Alcohols

Primary Alcohol + $KMnO_4$ or $K_2Cr_2O_7$ → _______________ → _______________
acidified *acidified*
(Oxidising agent)

Secondary Alcohols + $KMnO_4$ or $K_2Cr_2O_7$ → _______________
acidified *acidified*
(Oxidising agent)

Tertiary Alcohols + $KMnO_4$ or $K_2Cr_2O_7$ → _______________
acidified *acidified*
(Oxidising agent)

Reactions of Esters

In acidic conditions: Ester + Water → ________________ + ________________

*Ester + Hydroxide → –R + ________________

* Soap making is an example of alkaline hydrolysis of an ester. See section on soaps in Chapter 5.

e.g.

CH_2OOCR
|
$CHOOCR$ + 3NaOH → ________________ + ________________
|
CH_2OOCR

4.8 EMPIRICAL FORMULA CALCULATIONS

Students need to be able to determine by calculation, the empirical and molecular formulae and the structural formula of a compound from the analysis of combustion or other data. When analysing data, the determination of a(n):

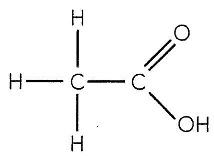
Empirical Formula: provides the simplest whole number ratio of moles of each element present

Molecular Formula: shows the actual whole number ratio of moles of each element present

Structural Formula: indicates how the atoms of each element are bonded to each other

e.g. Ethanoic Acid:

- empirical formula CH_2O
- molecular formula $C_2H_4O_2$
- structural formula is:



but this is often simplified to CH_3COOH.

Table 4.3 Compounds and their formulae.

Compound	Semi-structural formula	Molecular formula	Empirical formula
water	–	H_2O	H_2O
methane	–	CH_4	CH_4
ethane	CH_3CH_3	C_2H_6	CH_3
ethanoic acid	CH_3COOH	$C_2H_4O_2$	CH_2O
methyl methanoate	$HCOOCH_3$	$C_2H_4O_2$	CH_2O
* glucose			
* ethyne			
* benzene			

*(complete these)

Solving empirical formula problems is fairly straight forward. Just remember that we are trying to find the simplest mole ratio of the elements in the compound.

Empirical formula problems may either be:

- simple i.e. the masses of the elements in the compound (or %) are given.
- complex (one sample) – a single sample is burnt or reacted.
- complex (multi-sample) – two or more different sized samples are reacted.

Worked Example 4.3

Empirical Formula - 'simple type'.

A 12.00 g sample of a substance contains 6.54 g of carbon, 1.10 g of hydrogen while the remainder is oxygen.
(a) Find its empirical formula.
(b) If its molecular mass is 88.1, find its molecular formula.

		C	H	O
12.00 g sample	**Mass**	6.54 g	1.10 g	4.36 g
n =	**Moles**	6.54 / 12.01 = 0.5445	1.10 / 1.008 = 1.091	4.36 / 16.00 = 0.2725
to get this divide by smallest value (e.g. 0.2725)	**Mole ratio**	1.998	4.004	1.000
NB – do not round until the end	**Simple ratio**	2	4	1

(a) Hence empirical formula = C_2H_4O

(b) EF mass $= 2\,(12.01) + 4\,(1.008) + 16.00$
$= 44.05$

MF mass (given) $= 88.1$
$\therefore$ MF $= 2\,(\text{EF}) = C_4H_8O_2$

Question 4.30

A compound was found to contain 40.0% carbon, 6.67% hydrogen and 53.3% oxygen. Its relative molecular mass is 60.0. Find its empirical formula and molecular formula.

	C	H	O
Mass			
Moles			
Mole ratio			
Simple ratio			

Hint: For % problems assume that you have a 100 g sample

Hence empirical formula = ____________________

Also EF mass = ____________________

MF mass = ____________________

Worked Example 4.4

Empirical Formula - 'complex type'.

An organic compound was known to contain only carbon, hydrogen and oxygen. A 6.36 g sample was burnt and the products formed collected (3.01 g of water and 7.36 g of carbon dioxide).

(a) Determine the empirical formula.

(b) Find the molecular formula if the molecular mass is 228 g.

$$C?H?O? \quad + \quad O_2 \quad \rightarrow \quad \underset{7.36\ g}{CO_2} \quad + \quad \underset{3.01\ g}{H_2O}$$

Step 1 Find the masses of C, H, O in the original sample.

$\frac{m}{M}$

$$\text{mass of C} = \frac{12.01}{44.01} \times 7.36 = 2.008 \text{ g}$$

$$\text{mass of H} = \frac{(2)(1.008)}{18.016} \times 3.01 = 0.337 \text{ g}$$

$$\therefore \text{mass of O} = 6.36 - 2.008 - 0.337 = 4.015 \text{ g}$$

We now have a 'simple' Empirical Formula problem. Complete this example:

	C	H	O
Mass			
Moles			
Mole ratio			
Simple ratio			

Hence empirical formula = ____________________

Also EF mass = ____________________

MF mass = ____________________

Worked Example 4.5

Empirical Formula - 'complex type' - two samples, same mass.

A 0.6578 sample of a compound which was known to contain only C, H, O and S was completely burnt in excess oxygen. This produced 0.7413 g of carbon dioxide and 0.4551 g of water. The sulfur present was oxidised to the sulfate and precipitated as barium sulfate - 1.966 g being recovered.

(a) Determine the empirical formula.

Sample 1	C?H?O?S? 0.6578 g	+	O_2	→	CO_2 0.7413 g	+	H_2O 0.4551 g	+ ?
Sample 2	C?H?O?S? 0.6578 g	+	SO_4^{2-}	→	$BaSO_4$ 1.966 g.			

Note: the sulfur in the $BaSO_4$ all came from the original 0.6578 sample.

(b) A sample of the compound was vapourised and found to have a density of 3.49 g L^{-1} at S.T.P. Determine the molecular formula.

(a) Step 1
Find the masses of C, H, O and S in the sample. Because the samples are the same size we can assume that the amount of sulfur in the sample 1 is the same as sample 2.

* *complete all calculations.*

* Sample 1 mass of C $= \frac{12.01}{44.01} \times 0.7413 =$

mass of H = ____________ × ____________ =

* Sample 2 mass of S = 32.07 × ____________ =

∴ mass of O = 0.6578 – (   ) – (   ) – (   ) = g

We now have a 'simple' Empirical Formula problem. Complete this example:

	C	H	S	O
Mass				
Moles				
Mole ratio				
Simple ratio				

Hence E.F. = ________________________________

(b) Determine molecular formula by first finding molecular mass.

mass of 1 L = 3.49 g

∴ mass of 22.41 L = ?

∴ molecular mass = (3.49) (22.41) = ____________

Now EF mass = ____________, MF mass = ____________

∴ MF = ________________________________

Worked Example 4.6

Empirical Formula - 'complex type' - two samples of different mass.

A compound containing carbon, hydrogen, nitrogen and oxygen was analysed as follows. First a 1.476 g sample was burnt and 2.320 g of CO_2 and 0.3168 g of H_2O were formed. A second sample (1.167 g) was treated to convert all the nitrogen into ammonia. The ammonia was titrated and found to be 0.01389 mol.

(a) Calculate the empirical formula of the compound.

(b) A third sample of the compound (2.520 g) was vapourised and occupied 336 mL at STP. Use this information to determine its molecular formula.

(c) Sketch a possible structure for this organic compound.

Note: this problem involves two different sized samples for finding the empirical formula - so it is best to tackle it by finding the % of each element within each sample - then we will have a 'simple' problem.

Sample 1

$$\underset{1.476\text{ g}}{C?H?N?O?} + O_2 \rightarrow \underset{2.320\text{ g}}{CO_2} + \underset{0.3168\text{ g}}{H_2O} + ?$$

Step 1 Find the % by mass of C, H in sample 1.

$$\text{mass of C} = \frac{12.01}{44.01} \times 2.320 = 0.6331\text{ g}$$

$$\therefore \%\text{ C in sample 1} = \frac{0.6331}{1.476} \times 100 = 42.89\%$$

$$\text{mass of H} = \frac{(2)(1.008)}{18.016} \times 0.368 = 0.337\text{ g}$$

$\therefore$ % H in sample 1

Sample 2

$$\underset{1.167\text{ g}}{C?H?N?O?} + ? \rightarrow \underset{0.01389\text{ g}}{NH_3} + ?$$

Step 2 Find the % by mass of N in sample 2.

$$n(NH_3) = 0.01389\text{ mol}$$

$$\therefore n(N) = 0.01389\text{ mol}$$

$$\therefore m(N) = (0.01389)(14.01) = 0.1946\text{ g}$$

$$\therefore \%\text{ N in sample 2} = \frac{0.1946}{1.167} \times 100 = 16.68\%$$

Step 3 Determine the % by mass of O by subtraction from 100%.

$$\therefore \%\text{ O} = 100.0 - 42.89 - 2.402 - 16.68$$

$$= 38.03\%$$

We now have a 'simple' EF problem. * *assume a 100 g sample – complete all calculations*

	C	H	N	O
Mass				
Moles				
Mole ratio				
Simple ratio				

(a) Hence E.F. = ______________________________

(b) To find the molecular formula we need to find the molecular mass.

2.520 g	occupy	336 mL	at STP
? g	occupy	22.41 L	at STP?

$\therefore$ molecular mass $= (2.520)(\frac{22410}{336}) = 168$ g

Now EF mass = ______________________________

MF mass = 168 g

$\therefore$ MF = ______________________________

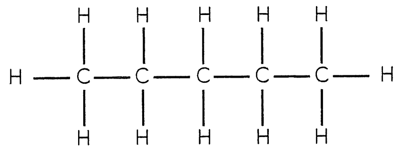
(c) Sketch a possible structure for this organic compound.

REVIEW QUESTIONS

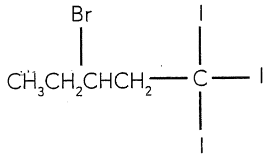
Chapter 4: Organic Chemistry

1. Use the IUPAC rules to name the following compounds.

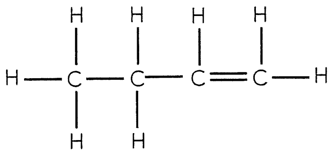
(a)



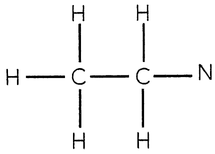
(b)



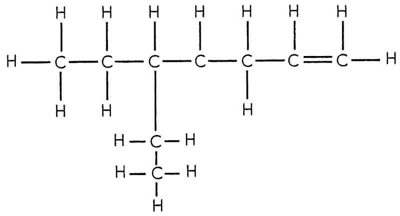
(c)



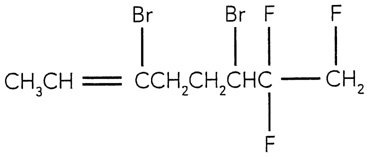
(d)



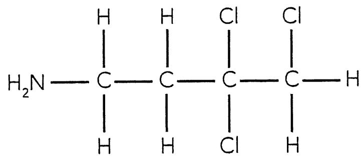
(e)



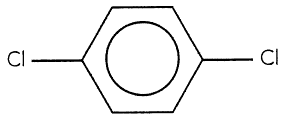
(f)



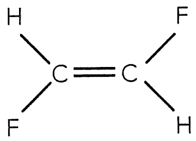
(g)



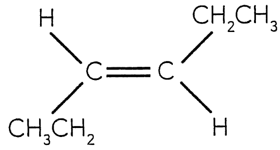
(h)



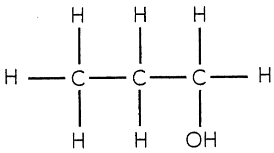
(i)



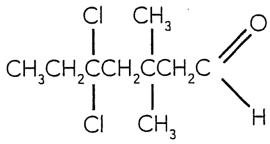
(j)



(k)



(l)



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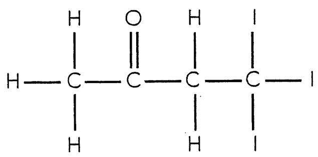

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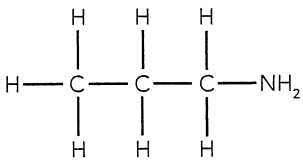

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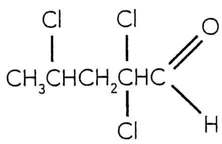

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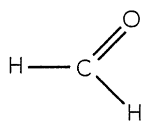
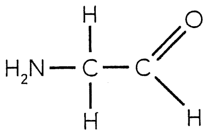


(q)



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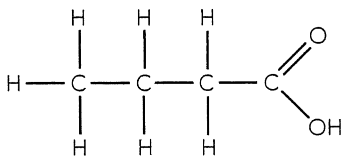
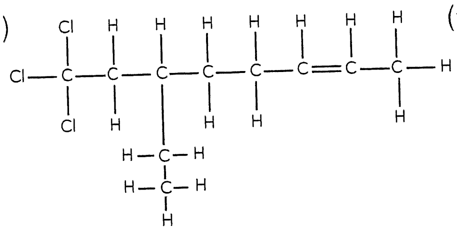
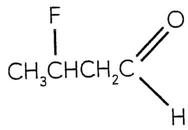
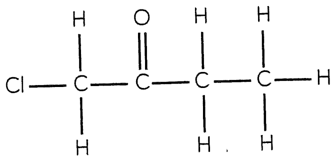
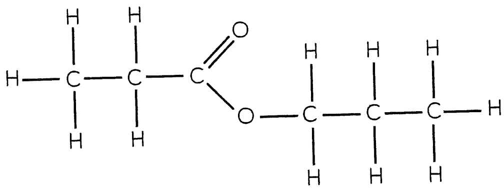
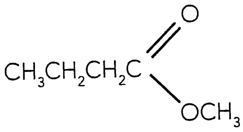
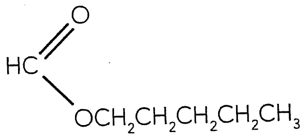


(s)  (t) 

(u)  (v) 

(w)  (x) 

2. Draw the structural formula of each of the following compounds.

(a) hexane
(b) 1,1,1-trichloroethane
(c) 2-methylbut-1-ene
(d) 4,4-diethylheptane
(e) heptan-1-amine
(f) 2-bromo-2-methylpropane
(g) 2-ethyl-3-methylpent-1-ene
(h) 2-chloro-3,3-diethyl-2,5-dimethylhexane
(i) *cis*-hept-3-ene
(j) *trans*-1,2-dibromobut-1-ene
(k) 4,4-dichloro-3-methylhexan-2-ol
(l) *trans*-1-chloropropene
(m) 4,5,5,5-tetrafluoropentan-2-one
(n) 3,4-diiodopentan-1-ol
(o) propanoic acid
(p) 3-hydroxybutanal
(q) *cis*-pent-2-ene
(r) 2,2,2-trichloroethanoic acid
(s) 1,2,3-tribromohexane
(t) 2,2-dichloropentan-1-amine
(u) 5-ethyl-5-methyloctan-4-one
(v) *trans*-1-chloropropene

3. Give a test and an observation that would permit the identification of each of the compounds in the pairs listed below.

(a) butanal and butanone
(b) butanal and hexane
(c) ethanol and propanone
(d) butan-1-ol and 2-methylpropan-2-ol
(e) methanol and pentan-1-ol

4. For each of the following redox reactions, write:

   (i) the oxidation and the reduction half equations

   (ii) the nett equation

   (iii) an expected observation.

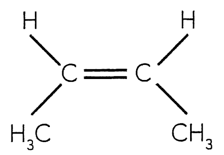
   (a) Acidified potassium dichromate is mixed with ethanol to form ethanal.

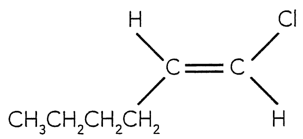
   (b) Acidified potassium permanganate is reacted with methanol to form methanoic acid.

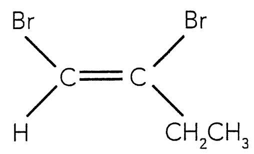
   (c) Acidified potassium permanganate is mixed with butan-2-ol.

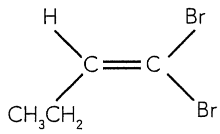
   (d) Acidified potassium permanganate is reacted with pentanal.

5. Name the following:

   (a) 

   (b) 

   (c) 

   (d) 

6. Write the balanced equation for the following reactions. Write the structural formula of any organic compound. (Assume correct reaction conditions are produced, e.g. catalyst present, UV light added, etc.)

   (a) propene + Br_2

   (b) hex-1-ene + Cl_2

   (c) methane + O_2

   (d) butane + Cl_2

   (e) hex-2-ene + HCl

   (f) octane + O_2

7. Name the type of substance whose properties are described below.

   (a) Fruity smelling compound that has a low solubility in water.

   (b) High melting point compared to other organic compounds of similar molar mass; ionise to some extent when dissolved in water.

   (c) Very soluble in water, reacts with acidified $KMnO_4$ and with carboxylic acids.

   (d) Not soluble in water; decolourises bromine water; can be used to form polymers.

8. Name the esters produced in the reactions below.

(a) methanol + methanoic acid

(b) butan-1-ol + ethanoic acid

(c) heptanoic acid + hexan-1-ol

(d) pentanoic acid + ethanol

9. Write the balanced equations for the production of the following esters.

(a) butyl heptanoate

(b) methyl pentanoate

(c) ethyl methanoate

(d) ethyl ethanoate

10. Esters can undergo hydrolysis. In acidic conditions a carboxylic acid and an alcohol are formed while in basic conditions an alcohol and a salt of a carboxylic acid will form.

(a) Complete the following equations for the hydrolysis of an ester.

(i) $CH_3CH_2COOCH_3 + H_2O \rightarrow$ (in acidic conditions)

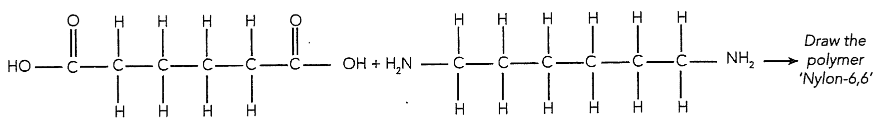
(ii) $CH_3CH_2COOCH_3 + OH^- \rightarrow$ (in basic conditions)

(b) What common household substance is produced by the hydrolysis of a complex ester in basic conditions? (See section 5.1.)

11. Dicarboxylic acids contain two carboxylic acids on the one hydrocarbon chain. Some dicarboxylic acids can be reacted with dialkanamines or dialkanols to produce polymers.

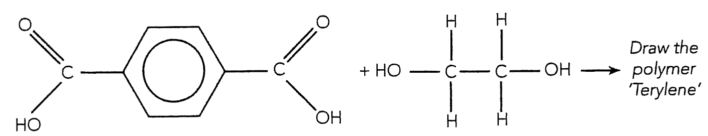
For each example below, draw a unit of the polymer that will form. The other product in each of the examples below is water.

(a)



hexanedoic acid hexan-1,6-diamine

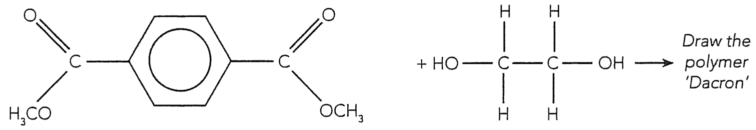
(b)



terephthalic acid ethan-1,2-diol

A third polymer can be made from the ester derivative of a dicarboxylic acid; e.g.

(c)

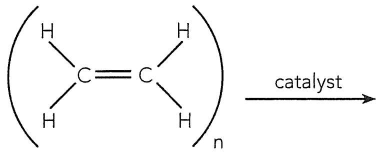


dimethyl terephthalate ethan-1,2-diol

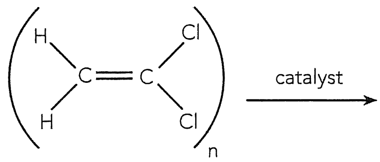
(d) Parts (a) and (b) show the formation of condensation polymers. Apart from the polymer, what is the other product formed?

12. Complete the following equations to show the formation of an addition polymer from an ethene based monomer.

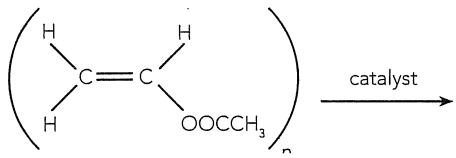
(a)



(b)



(c)



13. An organic compound was found to be a liquid at room temperature and composed of carbon, hydrogen and oxygen only. Complete combustion of a 9.994 g sample of the compound produced 14.65 g of carbon dioxide and 5.997 g of water.

A set of chemical tests on a different sample of the compound found that 2.22 g of the compound required 37.5 mL of 0.986 mol L^{-1} NaOH for neutralisation.

(a) Determine the empirical formula of the compound.
(b) Assuming the compound is monoprotic determine its molecular formula.
(c) Suggest a likely structural formula for the compound.

14. Qualitative analysis of an organic compound showed that it contained carbon, hydrogen and oxygen only.

When 0.392 g of this sample was burnt in excess oxygen it produced 0.320 g of water and 547 mL of carbon dioxide at 100.0°C and 101.3 kPa pressure.

When another 0.392 of the sample was heated to 250.0°C, the vaporised sample was found to create a pressure of 194 kPa in a 1.00×10^2 mL container.

(a) Determine the empirical formula of the compound.

(b) Determine the molecular formula of the compound.

(c) When this compound was reacted with methanol it was found to produce a liquid that smelt like apples.
(i) Draw the structural formula of the compound.
(ii) Name the product formed when it reacts with methanol.

15. A 5.945 g sample of a substance was found to be composed of 25.45% copper, 12.84% sulfur, 57.67% oxygen and 4.04% hydrogen.

(a) Determine its empirical formula.Further analysis of another 5.945 g of the sample found that when converted from its normal hydrated form to the anhydrous form a weight loss of 2.145 g occurred.

(b) Determine the molecular of the compound showing the water of crystallisation.

16. An organic compound containing oxygen, carbon and hydrogen was labelled X and was subject to the following set of experiments:

I. Substance X was found to be oxidised by potassium permanganate to produce substance Y.

II. Substance X reacted with substance Y to produce a sweet smelling substance labelled substance Z.

III. Substance X and substance Y reacted with sodium.

IV. Substance X was found to be a straight chain compound containing less than 5 carbon atoms.

V. Substance Z underwent hydrolysis in acidic conditions to produce substances X and Y.

VI. 11.62 grams of substance Z was burnt in oxygen and produced 26.41 g of carbon dioxide and 10.81 g of water.

(a) Determine the empirical formula of substance Z.

(b) Draw a possible structural formula of substance Z.

17. When 3.990 g of an organic compound, containing only C, H and N, was burnt in excess oxygen, 8.927 g of carbon dioxide, 5.478 g of water vapour and 3.111 g of nitrogen dioxide were produced.

(a) Determine the empirical formula of the compound.

When a further 3.990 g sample of the compound was vapourised and collected at 210.0°C and 101.3 kPa pressure, it was found to occupy a volume of 2.680 L.

(b) Calculate the molecular formula of the compound.

(c) Given that the only functional group was in a "primary" location draw the structural formula of the compound and give its IUPAC name.

CHP 4: ORGANIC CHEMISTRY

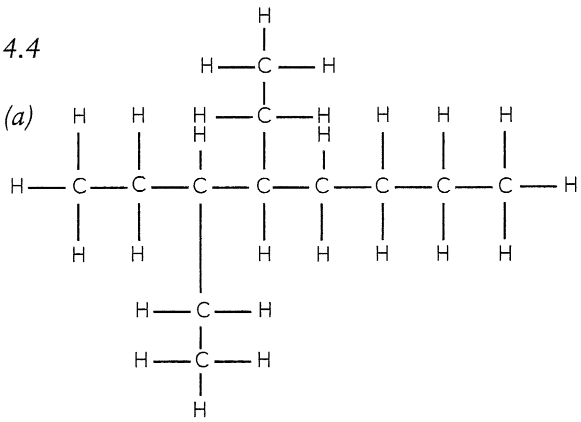
Chapter Questions

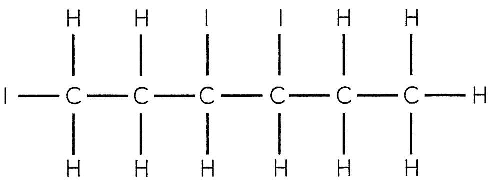
4.1 (a) *alkene (=)* (b) *alkene (=)*
(c) *-oic acid(-COOH)* (d) *halide (-Cl)*
(e) *ester* $-\overset{O}{\overset{\|}{C}}-O-$ (f) *ketone* $-\overset{O}{\overset{\|}{C}}-$
(g) *secondary alcohol (–OH)*
(h) *amine* ($-NH_2$)

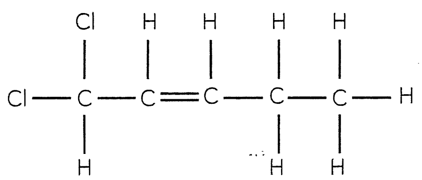
4.2

3	propene	propyl
4	1-butene	butyl
5	2-pentene	pentyl
6	1-hexene	hexyl
7	3-heptene	heptyl
8	3-octene	octyl

4.3
(a) *butane*
(b) *but-1-ene*
(c) *chloromethane*
(d) *3-methylbut-1-ene*
(e) *4-ethyl-2-methylheptane*
(f) *3-methylpentane*
(g) *1,2-difluoro-2-methylhex-3-ene*
(h) *4-chloro-1,3,4-trifluorobut-1-ene*

4.4
(a)


(b)


(c)


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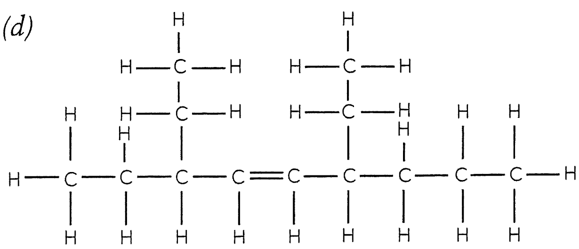
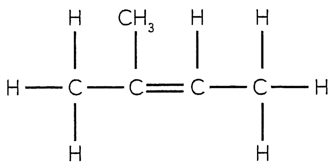
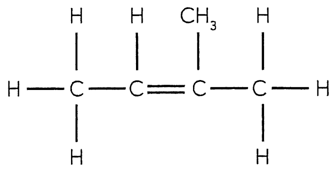
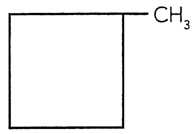


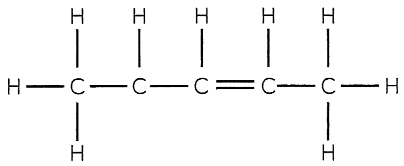
4.5
(a) *incorrect numbering – should be but-2-ene*
(b) *longest chain is butane not propene*
(c) *incorrect numbering – should be pent-1-ene*
(d) *incorrect numbering – should be 2,3-dimethylpent-1-ene*

4.6
(a)


(b)


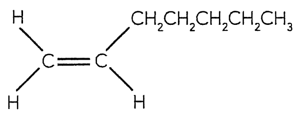
Not an isomer – incorrectly numbered – same as (a).

(c)


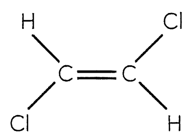
(d)


Wrong name - pent-2-ene - so same as Isomer 2.

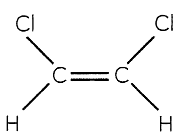
4.7 (a) cis-*3-chlorohex-3-ene*
(b) cis-*2,3-dichlorobut-2-ene*
(c) trans-*1,4-dichlorobut-2-ene*
(d) cis-*1,2-dibromobut-1-ene*

4.8
(a)


Not a geometric isomer.

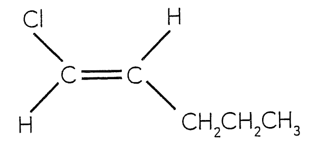
(b)


trans

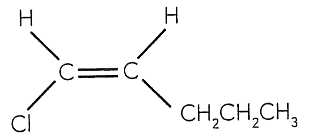


cis

(c)



trans



cis

4.9

(a) *3-methylpentan-1-ol*

(b) *5-bromo-1,1,1-trifluoro-3,4-dimethylpentan-2-ol*

4.10

(a) *heptan-2-ol*

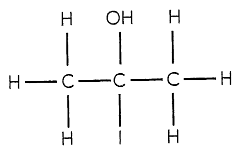
(b) *3-chloro-1,1-diiodopentan-2-ol*

(c) *5,5,5-trichloro-3-methylpentan-1-ol*

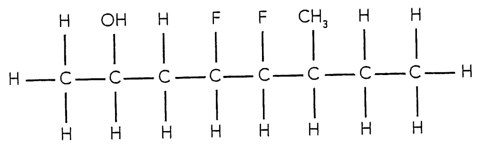
(d) *2-chloro-2,3-dimethylbutan-1-ol*

4.11

(a)



(b)



4.12

(a) $(CH_3CH_2CH_2CH_2OH \rightarrow$

$CH_3CH_2CH_2C\genfrac{}{}{0pt}{}{/\!/O}{\backslash H} + 2H^+ + 2e^-) \times 3$

$Cr_2O_7 + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$3CH_3CH_2CH_2CH_2OH + Cr_2O_7^{2-} + 8H^+ \rightarrow$

$3CH_3CH_2CH_2C\genfrac{}{}{0pt}{}{/\!/O}{\backslash H} + 7H_2O + 2Cr^{3+}$

(b) $(CH_3CH_2\overset{OH}{\overset{|}{C}}HCH_2CH_3 \rightarrow$

$CH_3CH_2\overset{O}{\overset{\|}{C}}CH_2CH_3 + 2H^+ + 2e^-) \times 5$

$(MnO_4^- + 8H^+ + 5e^- \rightarrow$

$Mn^{2+} + 4H_2O) \times 2$

$5CH_3CH_2\overset{OH}{\overset{\|}{C}}CH_2CH_3 + 2MnO_4^- + 6H^+ \rightarrow$

$5CH_3CH_2\overset{O}{\overset{\|}{C}}CH_2CH_3 + 2Mn^{2+} + 8H_2O$

(c) $C_4H_{10}O + 6O_2 \rightarrow 4CO_2 + 5H_2O$

4.13

(a) *3,5-dibromo-5-iodopentanal*

(b) *3,3-dimethylbutanone*

(c) *3,3,3-trichloropropanoic acid*

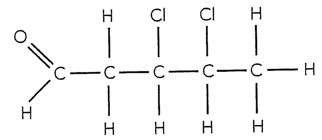
(d) *3-chloro-3-methylbutanal*

(e) *4,4,4-trifluoro-3,3-diiodobutanone*

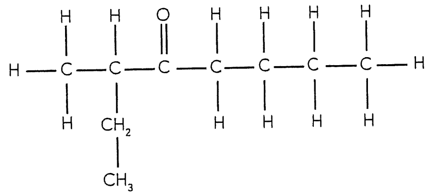
(f) *3-methylheptanal*

4.14

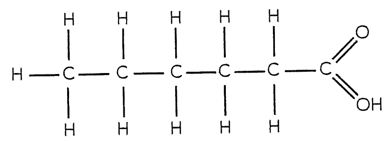
(a)



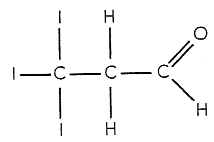
(b)



(c)

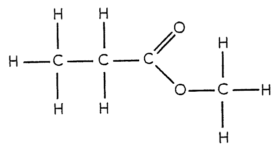


(d)

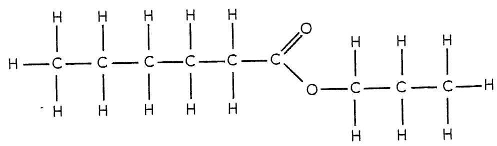


4.15

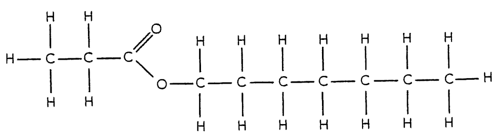
(a) *methyl propanoate*



(b) *propyl hexanoate*



(c) *heptyl ethanoate*



4.16

(a) *ethanol + propanoic acid*

(b) *1-octanol + butanoic acid*

(c) *1-propanol + methanoic acid*

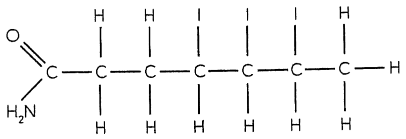
4.17

(a) *7,7,7-trifluoro-3-methylheptanamide*

(b) *2,5-dibromohexanamide*

4.18

(a)



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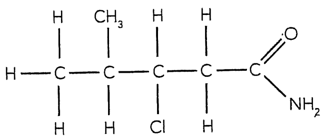


4.19

(a) *2-methylpentan-1-amine*

(b) *2,5-dichloro-3,4-dimethylhexan-1-amine*

4.20

Attached chain is 'H' – non-polar and neutral

4.21

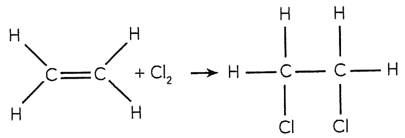
(a) $H_2N\text{-}CH_2\text{-}COOH \rightarrow {+}H_3N\text{-}CH_2COO\text{-}_{(aq)}$ *(zwitterion form)*

(b) $+H_3N\text{-}CH_2\text{-}COO\text{-}_{(aq)} + OH\text{-}_{(aq)} \rightarrow H_2N\text{-}CH_2\text{-}COO\text{-}_{(aq)} + H_2O_{(l)}$

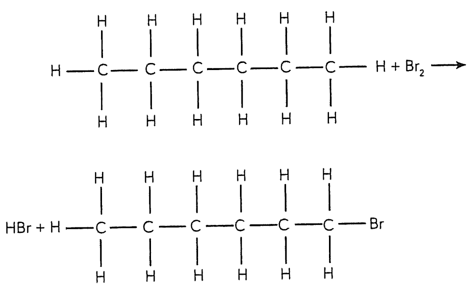
(c) $+H_3N\text{-}CH_2\text{-}COO\text{-}_{(aq)} + H+_{(aq)} \rightarrow {+}H_3N\text{-}CH_2\text{-}COOH_{(aq)}$

4.22

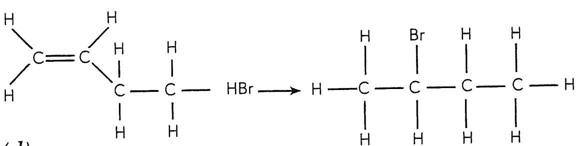
(a)



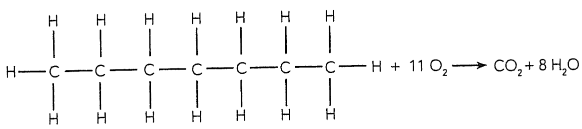
(b)



(c)

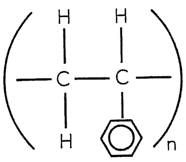


(d)



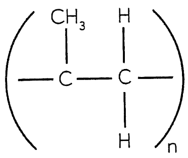
4.23

(a)



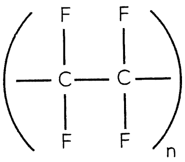
polymer = polystyrene

(c)

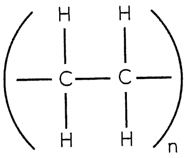


polymer = polypropene

(b)



(d)



polymer = polyethene or polythene

4.24 *4 monomer units drawn.*

4.25

Motor oil is non-polar - use a non-polar solvent, such as petrol, to dissolve it. Use a detergent to clean up residue.

4.26

Methylated spirits is polar and so will not be a very good solvent for non-polar materials. Tends not to leave streaks because it evaporates so quickly.

4.27

See next page.

4.28

Primary alcohol + $KMnO_4$	aldehyde or carboxylic acid	loss of purple colour, odour produced
Secondary alcohol + $K_2Cr_2O_7$	ketone	orange solution turns green, odour produced
Tertiary alcohol + $KMnO_4$	NR	NR
Alkane + Cl_2	chloroalkane	faint yellow/ green colour disappears
Alkene + Cl_2	dichloroalkane	as above

4.29

COMBUSTION
Hydrocarbons
Alcohols
Aldehydes, ketones
Esters
Carboxylic acids

$+ O_2 \rightarrow CO_2 + H_2O$

SUBSTITUTION

Alkane + Halogen → haloalkane + hydrogen halide

Alkene + Halogen → dihaloalkane

4.27

Func. Group	Gen. Formula	Name ends in	Solubility	Produced by
–Cl –I –Br	R – X	has the prefix matching the alkene	usually low - increases as molecule becomes more polar	substitution –C–C– + Br_2 → –C–C– Br + HBr addition C=C + Br_2 → –C–C–
>C=C< (alkene)	R1–C=C–R_2	–ene	not soluble	not applicable
–NH_2 (primary amine)	R–C–NH_2	–amine	soluble	not applicable
–OH (alcohols)	R–OH	–ol	soluble	hydration of an alkene
–CHO (aldehydes)	R–CHO	–al	soluble	oxidation of a primary alcohol
O ‖ –C– (ketones)	R_1–C–R_2 O	–one	soluble	oxidation of a secondary alcohol
–COOH (carboxylic acids)	R–C	–oic acid	soluble	oxidation of a primary alcohol or of an aldehyde
O ‖ –C–O – (esters)	R–C	–oate	not soluble	alcohol and carboxylic acid

4.29 [continued]

OXIDATION OF ALCOHOLS

Primary Alcohol + $KMnO_4$ or $K_2Cr_2O_7$ $\xrightarrow{\text{Oxidising agent}}$ aldehyde → carboxylic acid

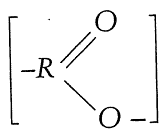
Secondary Alcohols + $KMnO_4$ or $K_2Cr_2O_7$ $\xrightarrow{\text{Oxidising agent}}$ ketone

Tertiary Alcohols + $KMnO_4$ or $K_2Cr_2O_7$ $\xrightarrow{\text{Oxidising agent}}$ no reaction

REACTIONS OF ESTERS

IN ACIDIC CONDITIONS:

Ester + Water → carboxylic acid + alcohol

** Ester + Hydroxide →*  *+ alcohol*

** Soap making is an example of the hydrolysis of an ester.*

e.g. CH_2OOCR
|
$CHOOCR$ + $3NaOH$ → *soap + glycerol*
|
CH_2OOCR

Table 4.3

glucose	$C_6H_{12}O_6$	$C_6H_{12}O_6$	CH_2O
ethyne	CHCH	C_2H_2	CH
benzene	C_6H_6	C_6H_6	CH

4.30

	C	H	O
mass	40.0	6.67	53.3
moles	40.0/12.01 = 3.33	6.67/1.008 = 6.62	53.5/16.00 = 3.33
mole ratio	1	2	1
simple ratio	1	2	1

Empirical formula = CH_2O
EF mass = 30
MF mass = 60

$\therefore$ *MF mass* = 2 × *EF mass*
$\therefore$ *MF* = 2 × *EF* = $C_2H_4O_2$

Worked Example 4.4

	C	H	O
mass	2.008	0.337	4.015
moles	2.008/12.01 = 0.167	0.337/1.008 = 0.334	4.015/16.00 = 0.251
mole ratio	1	1.9996	1.5009
simple ratio	2	4	3

EF = $C_2H_4O_3$
EF mass = 76
MF mass = 228
$\therefore$ *MF* = 3 × *EF* = $C_6H_{12}O_9$

Worked Example 4.5

(a) *sample 1* *mass of H* =
= 0.05093 *g*
sample 2 *mass of S* =
= 0.2702 *g*
$\therefore$ *mass of O*
= 0.6578 – 0.2023 – 0.05093 – 0.2702
= 0.1344 *g*

	C	H	S	O
mass	0.2023	0.05093	0.2702	0.1344
moles	0.01684	0.05052	0.00842	0.00840
mole ratio	2.005	6.014	1.003	1
simple ratio	2	6	1	1

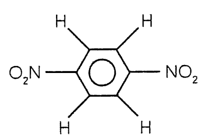
Hence EF = C_2H_6SO

(b) *molecular mass* = 78.21 *g*
EF mass = 78.14, *MF mass* = *EF mass*
MF = C_2H_6SO

Worked Example 4.6

	C	H	N	O
moles	3.571	2.383	1.1906	2.377
mole ratio	3.000	2.001	1	1.996
simple ratio	3	2	1	2

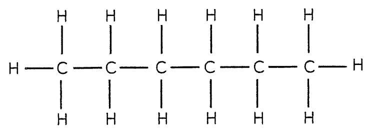
(a) *EF* = $C_3H_2NO_2$
(b) *EF mass* = 84
$\therefore$ *MF* = 2 × *EF* = $C_6H_4N_2O_4$
(c)



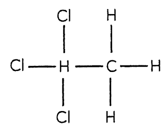
4. Review Questions

1. (a) *pentane*
(b) *3-bromo-1,1,1-triiodopentane*
(c) *but-1-ene*
(d) *ethanamine*
(e) *4-ethylhept-1-ene*
(f) *3,6-dibromo-7,7,8-trifluorooct-2-ene*
(g) *3,3,4-trichlorobutan-1-amine*
(h) *1,4-dichlorobenzene*
(i) trans*-1,2-difluoroethene*
(j) trans*-hex-3-ene*
(k) *propan-1-ol*
(l) *5,5-dichloro-3,3-dimethylheptanal*
(m) *4,4,4-triiodobutanone*
(n) *propan-1-amine*
(o) *2,2,4-trichloropentanal*
(p) *methanal*
(q) *α-amino acid (glycine)*
(r) *butanoic acid*
(s) *8,8,8-trichloro-6-ethyl-5-methyloct-2-ene*
(t) *3-fluorobutanal*
(u) *1-chlorobutanal*
(v) *propyl propanoate*
(w) *methyl butanoate*
(x) *pentyl methanoate*

2. (a)



(b)



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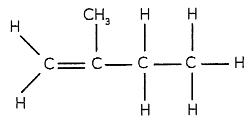

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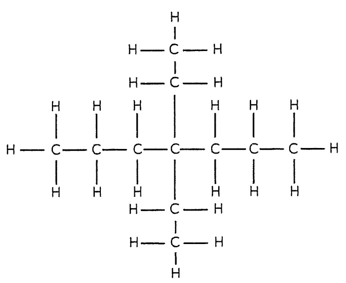

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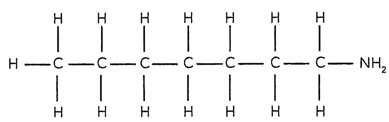
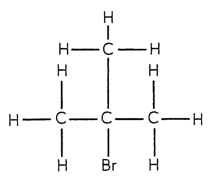


(f)



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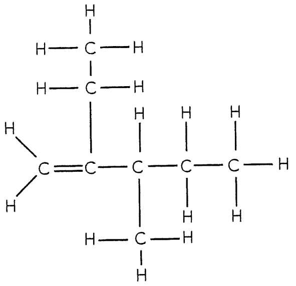

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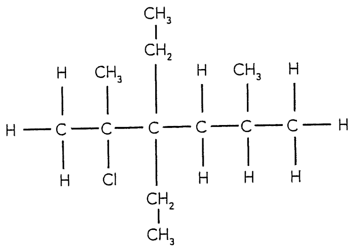
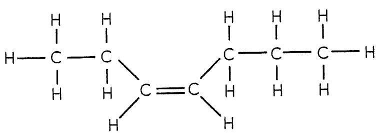


(i)



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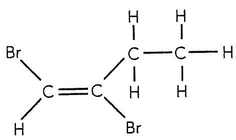

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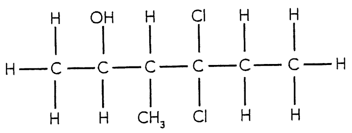

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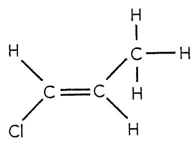

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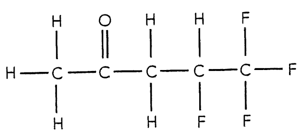

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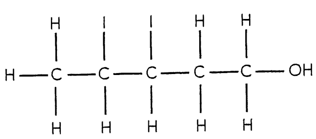

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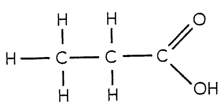
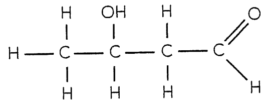


(p)



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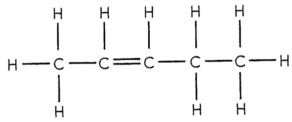

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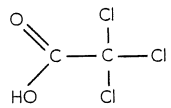

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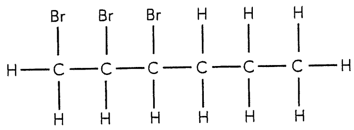

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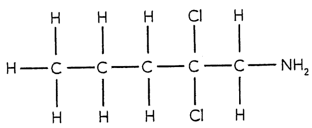
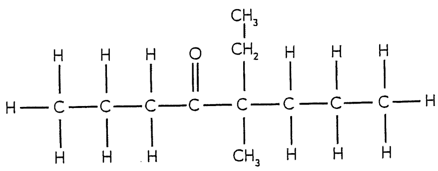


(u)



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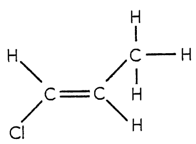
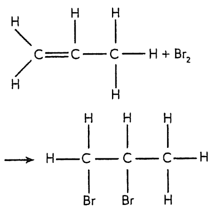
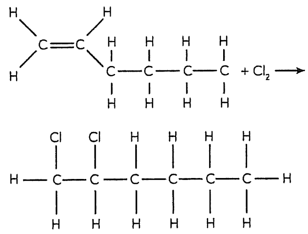
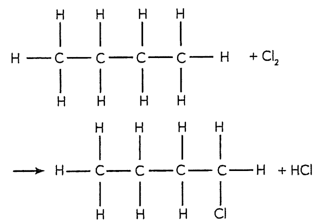
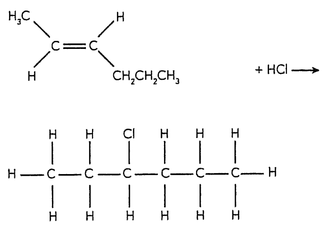


3.
(a) *Add several drops of acidified $KMnO_4$ solution to both. The butanal will decolourise the $KMnO_4$ while the butanone will not*
(b) *Add several drops of acidified $KMnO_4$ solution to both. The butanal will decolourise the $KMnO_4$ while the hexane will not (or check solubility in water - butanal is more soluble than hexane).*
(c) *As (c) to (e) above - ethanol decolourises $KMnO_4$ while propanone does not.*
(d) *Add a small piece of sodium to both. The sodium reacts more vigorously with 1-butanol than it does with 2-methylpropan-2-ol.*
(f) *Check solubilities, methanol is much more soluble in water than 1-pentanol. (Check boiling points - methanol has a lower boiling point than 1-pentanol.)*

4.
(a) (i) *oxidation*
$(CH_3CH_2OH \rightarrow CH_3CHO + 2H^+ + 2e^-) \times 3$
reduction
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
(ii) *nett*
$3CH_3CH_2OH + Cr_2O_7^{2-} + 8H^+$
$\rightarrow 3CH_3CHO + 2Cr^{3+} + 7H_2O$
(iii) *orange solution turns green when mixed with colourless solution*

(b) (i) *oxidation*
$(H_2O + CH_3OH \rightarrow HCOOH + 4H^+ + 4e^-) \times 5$
reduction
$(MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O) \times 4$
(ii) *nett*
$5CH_3OH + 4MnO_4^- + 12H^+$
$\rightarrow 5HCOOH + 4Mn^{2+} + 11H_2O$
(iii) *purple solution becomes colourless on mixing with colourless liquid*
(c) (i) *oxidation*
$(CH_3\underset{|\atop OH}{C}HCH_2CH_3 \rightarrow CH_3\underset{\|\atop O}{C}CH_2CH_3 + 2H^+ + 2e^-) \times 5$

reduction
$(MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O) \times 2$
(ii) *nett*
$5CH_3\underset{|\atop OH}{C}HCH_2CH_3 \rightarrow 2MnO_4^- + 6H^+$
$\rightarrow 5CH_3\underset{\|\atop O}{C}CH_2CH_3 \rightarrow 2Mn^{2+} + 8H_2O$
(iii) *purple solution becomes colourless on mixing with colourless liquid*
(d) (i) *oxidation*
$(CH_3CH_2CH_2CH_2CHO + H_2O$
$\rightarrow CH_3CH_2CH_2CH_2COOH + 2H^+ + 2e^-) \times 5$
reduction
$(MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O) \times 2$
(ii) *nett*
$5CH_3CH_2CH_2CH_2CHO + 2MnO_4^- + 6H^+$
$\rightarrow 5CH_3CH_2CH_2CH_2COOH + 2Mn^{2+} + 3H_2O$
(iii) *purple solution becomes colourless on mixing with colourless liquid*

5. (a) cis-*but-2-ene*
(b) trans-*1-chorohex-l-ene*
(c) cis-*1,2-dibromobut-l-ene*
(d) *1,1-dibromobut-l-ene*
6. (a)

(b)

(c) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
(d)

(e)

(f) $2C_8H_{18} + 17O_2 \rightarrow 8CO_2 + 18H_2O$

7. (a) *ester*
(b) *carboxylic acid*
(c) *alcohol*
(d) *alkene*

8. (a) *methyl methanoate*
(b) *butyl ethanoate*
(c) *hexyl heptanoate*
(d) *ethyl pentanoate*

9. (a) $CH_3CH_2CH_2CH_2OH + C_6H_{13}COOH$
$\rightarrow C_6H_{13}C{\overset{\displaystyle //O}{\underset{\displaystyle \backslash O-CH_2CH_2CH_2CH_3}{}}}$

(b) $CH_3OH + C_4H_9COOH$

$$\rightarrow C_4H_9C{\overset{\displaystyle O}{\diagup\!\!\diagup}} \atop {\diagdown\ O - CH_3}$$

(c) $CH_3CH_2OH + HCOOH$

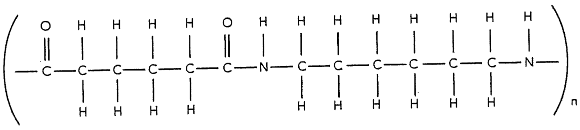
$$\rightarrow H - C{\overset{\displaystyle O}{\diagup\!\!\diagup}} \atop {\diagdown\ O - CH_2CH_3}$$

(d) $CH_3CH_2OH + CH_3COOH$

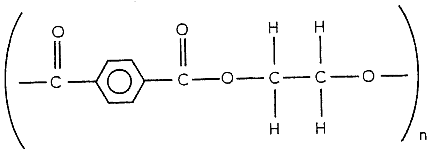
$$\rightarrow CH_3C{\overset{\displaystyle O}{\diagup\!\!\diagup}} \atop {\diagdown\ O - CH_2CH_3}$$

10.
(a)
(i) $CH_3CH_2COOCH_3 + H_2O$
$\rightarrow CH_3CH_2COOH + CH_3OH$
(ii) $CH_3CH_2COOCH_3 + OH^-$
$\rightarrow CH_3CH_2COO^- + CH_3OH$
(b) Soap.

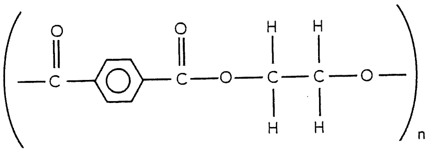
11.
(a) Nylon "6,6"



(b) Terylene

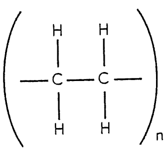


(c) Dacron

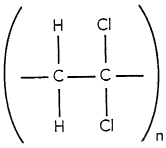


(d) H_2O

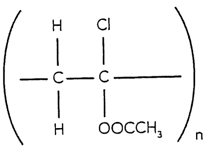
12. (a)



(b)



(c)



13. $\underset{9.994\ g}{C_xH_yO_z} + O_2 \rightarrow \underset{14.65\ g}{CO_2} + \underset{5.979\ g}{H_2O}$

(a) $n(C) = n(CO_2) = \dfrac{14.65}{44.01} = 0.3329\ mol$

$m(C) = 0.3329 \times 12.01 = 3.998\ g$

$n(H) = 2n(H_2O) = \dfrac{2 \times 5.997}{18.016}$

$= 0.6657\ mol$

$m(H) = 0.6657 \times 1.008 = 0.6711\ g$
$m(O) = m(sample) - [m(C) + m(H)]$
$= 9.994 - [3.998 + 0.6711] = 5.325\ g$
$n(O) = \dfrac{m}{M} = 0.3329$

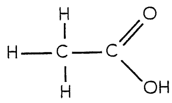
C	H	O
0.3329	0.6657	0.3329
1	2	1

EF = CH_2O

(b) $H + OH^- \rightarrow H_2O$
$n(NaOH) = cV = 0.986 \times 0.0375$
$= 0.0370$
$\therefore n(H^+) = 0.0370$
If monoprotic then n(compound)
$= n(H^+) = 0.0370$
$M(compound) = \dfrac{m}{M} = \dfrac{2.22}{0.0370}$

$= 60.04\ g\ mol^{-1}$
mass of EF = 30, MF = 2 × EF
Molecular formula = $C_2H_4O_2$

(c)



(ethanoic acid)

14. $\underset{0.392\ g}{C_xH_yO_z} + O_2 \rightarrow CO_2 + \underset{0.320\ g}{H_2O}$

(a) $n(C) = n(CO_2) = \dfrac{PV}{RT} = \dfrac{101.3 \times 0.547}{8.315 \times 373.1}$

$= 1.79 \times 10^{-2}\ mol$
$m(C) = 1.79 \times 10^{-2} \times 12.01 = 0.215\ g$
$n(H) = 2n(H_2O) = \dfrac{2 \times m}{M} = \dfrac{2 \times 0.320}{18.016}$

$= 3.55 \times 10^{-2}$
$m(H) = 3.58 \times 10^{-2}\ g\ mol$
$m(O) = m(sample) - [m(C) + m(H)]$
$= 0.392 - [0.215 + 0.0358] = 0.141\ g$
$n(O) = \dfrac{m}{M} = 8.84 \times 10^{-3}$

C	H	O
$\dfrac{0.0179}{0.00884}$	$\dfrac{0.0355}{0.00884}$	$\dfrac{0.00884}{0.00884}$
2.02	4.02	1

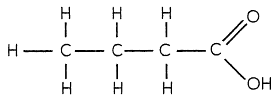
$\therefore$ EF = C_2H_4O

(b) $n = \frac{PV}{RT} = \frac{194 \times 0.100}{8.315 \times 523.1} = 4.46 \times 10^{-3}\ mol$

$M = \frac{m}{n} = \frac{0.392}{4.46 \times 10^{-3}} = 87.9\ g\ mol^{-1}$

mass of EF = 44,
mass of MF = 2 × mass of EF
$\therefore$ *Molecular formula =* $C_4H_8O_2$

(c) (i)



(ii) *methyl butanoate*

15. (a)

	Cu	S	O	H
mass	5.945 × 25.45%	5.945 × 12.84%	5.945 × 57.67%	5.945 × 4.04%
moles	$\frac{1.513\ g}{63.55}$	$\frac{0.7633 g}{32.07}$	$\frac{3.428\ g}{16.00}$	$\frac{0.240\ g}{1.008}$
	$\frac{0.02381}{0.02380}$	$\frac{0.02380}{0.02380}$	$\frac{0.2142}{0.02380}$	$\frac{0.2383}{0.02380}$
	1	1	9	10

$EF = CuSO_9H_{10}$

(b) $m(H_2O)$ lost = 2.145 g
$n(H_2O)$ lost = = = 0.1191 mol
mole ratio of H_2O *lost =* =5.00
$MF = CuSO_4.5H_2O$

16.

(a) $C_xH_yO_z + O_2 \rightarrow CO_2 + H_2O$

11.61 g 26.41 g 10.81 g

$n(C) = \frac{26.41}{44.01} = 0.6001\ mol$

$m(C) = 0.6001 \times 12.01 = 7.207\ g$

$n(H) = \frac{2 \times 10.81}{18.016} = 1.200\ mol$

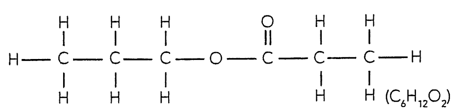
$m(H) = 1.200 \times 1.008 = 1.2096\ g$
$m(O) = 11.61 - (7.207 + 1.2096)$
$= 3.193\ g$

$n(O) = \frac{3.193}{16.00} = 0.1996\ mol$

C	H	O
$\frac{0.6001}{0.1996}$	$\frac{1.200}{0.1996}$	$\frac{0.1996}{0.1996}$
3.006	6.01	1

$EF = C_3H_6O$

(b) *Information suggests Z is an ester made from an alcohol and carboxylic acid of equal length and less than 5 carbons each.*
i.e.



17.

(a) $C_xH_yN_z + O_2 \rightarrow CO_2 + H_2O + NO_2$

3.990 g ~~8.927 g 5.478 g~~ 3.111 g

$n(C) = \frac{8.927}{44.01} = 0.2028\ mol$

$n(H) = \frac{2 \times 5.478}{18.016} = 0.6081\ mol$

$n(N) = \frac{3.111}{46.01} = 0.0676\ mol$

C	H	N
$\frac{0.2028}{0.0676}$	$\frac{0.6081}{0.0676}$	$\frac{0.0676}{0.0676}$
2.999	8.993	1

$EF = C_3H_9N$

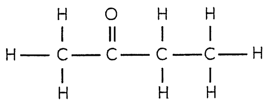
(b) $n = \frac{PV}{RT} = \frac{101.3 \times 2.680}{8.315 \times 483.1} = 0.0676$

$M = \frac{m}{n} = \frac{3.990}{0.0676} = 58.99$

molar mass of EF = 59 = molar mass of MF

$\therefore$ *Molecular formula =* C_3H_9N

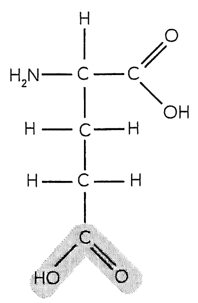
(c)



propanamine

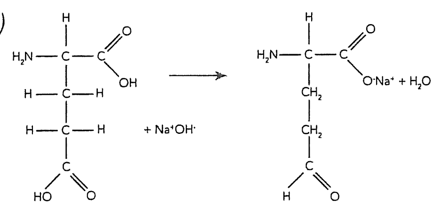
18. FOR THE EXPERTS

(a)

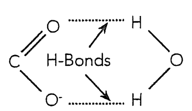


(b) *acidic COOH group, polar.*

(c)



(d) *Soluble in* H_2O *owing to*



and